

Experimental Design for Policy Choice

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We show how to optimally design experiments when the resulting data will be used to choose a welfare-maximizing policy subject to constraints. A decision maker seeks to maximize Bayes expected welfare by choosing a policy whose effects depend on an unknown finite-dimensional parameter. The decision maker has access to a first wave of experimental data with a fixed design but may choose the design of a second wave that will be collected before choosing the policy. The resulting experimental design–policy choice problem involves a dynamic program with a very high-dimensional state and is generally intractable in finite samples. We propose a tractable approximation based on the limit experiment and show it is asymptotically optimal using a new asymptotic representation theorem for adaptive experiments with continuous treatments. We apply the method to a conditional cash transfer experiment and demonstrate the potential for large gains from tailoring the experiment to the policy choice.

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1 INTRODUCTION

Economists often use experiments to evaluate, compare, and choose policies. Given data from an experiment and an econometric model, decision makers can evaluate counterfactual policies and choose the one that maximizes some measure of welfare, subject to budgetary or other constraints. Experimental design can play an important role in this process, as better estimates of the effects of counterfactual policies can minimize the chances of choosing suboptimal policies. Given a finite experimental budget, some experimental designs will be more informative about the optimal policy than others. Such experiments should be preferred when the ultimate goal of the experiment is to inform policy choice.

Despite the fact that choosing policies is a ubiquitous objective for decision makers, very little is known about how to optimally design experiments for this purpose. Instead, much of the guidance on experimental design assumes the objective of maximizing the precision of parameter or treatment effect estimates (see S. Athey and Imbens (2017), as well as Section 1.1 below for a review). However, not all parameters are equally important for choosing policies. For example, estimating the average effect of a welfare benefits program may not be helpful for choosing how benefits should vary based on a recipient’s income or number of dependents. Likewise, precisely estimating heterogeneous effects across variables that the policy cannot target will not generally improve the performance of the chosen policy. Instead, the experiment should focus on learning about the effects of counterfactual policies on welfare, prioritizing policies that are more likely to be welfare-enhancing. Because the objective and constraints of a policy choice problem differ from those of a parameter estimation problem, an experiment designed around one objective may not be well-suited for the other.

This paper provides a method for designing experiments specifically for the purpose of informing policy choice. Given an econometric model, a welfare objective, and a set of feasible policies, our proposed method uses a first wave of experimental data to characterize which model parameters are most valuable to learn about for the purpose of policy choice, and then designs a second wave to optimally learn about these policy-relevant parameters. The resulting experiment delivers data which is maximally informative for choosing the best policy.

We consider a decision maker who must choose a policy to maximize expected welfare subject to budgetary or other constraints. The effects of counterfactual poli-

cies are described by a structural econometric model governed by an unknown finite-dimensional parameter. The decision maker has access to a first wave of experimental data and has the opportunity to design and run a second wave before choosing the policy. The resulting joint experimental design–policy choice problem is a dynamic decision problem which is difficult to solve due to the high-dimensionality of the state space. Since the policy depends on the realized data, a naive backward-induction approach requires the decision maker to specify which policy they would choose for every possible dataset that the experiment could generate. Even in simple settings, the state can consist of thousands of continuous state variables. As a practical matter, solving this problem directly is not possible.

We propose a tractable method for finding the optimal experiment using two approximations to the decision problem. First, we approximate the decision maker’s information environment by supposing they only observe a Gaussian estimate of model parameters from each wave. This reduces the question of selecting an experimental design to that of selecting a variance-covariance matrix for the Gaussian estimate. Second, we approximate the nonlinear welfare objective with a quadratic function. Under these two approximations, the decision maker’s problem of choosing a policy depends only on a Gaussian estimate of a particular linear combination of the model parameters—namely, the marginal effect of making small adjustments to the policy. This simplification enables us to develop an efficient algorithm to solve for the optimal experiment in practice.

We show that our proposed method is asymptotically optimal in the sense that a decision maker who (i) uses our method to select the experimental design, and (ii) uses the resulting data to choose a policy, will have the highest possible limiting welfare across all feasible experimental designs. We show this by validating the two approximations underlying our method in an asymptotic regime in which both waves of the experiment grow large. First, we establish a new asymptotic representation theorem for adaptive experiments with continuous treatments. This justifies the Gaussian information environment. Second, we show that the constrained policy choice problem is locally approximated by a quadratic program. This establishes that the value of this quadratic program in a Gaussian environment serves as an asymptotic bound on the performance of any experimental design–policy choice procedure in the finite-sample problem. Moreover, our proposed method achieves this bound, and hence achieves the same asymptotic welfare as directly solving the intractable finite-

sample decision problem.

We demonstrate the method in an application to the Progresa conditional cash transfer experiment (Schultz 2004, Gertler 2004). Taking the graduation rate of students as the welfare objective, we consider a decision maker who can offer a subsidy to students to encourage school attendance. The subsidy can vary by gender and grade of enrollment, but is subject to a budget constraint on the total amount of money offered over a student’s educational career. The decision maker has access to a first wave of experimental data from the Progresa experiment, which she uses to design a second wave of the experiment to learn about the optimal subsidy policy. In this application, we estimate that our experimental design reduces the regret of the downstream policy choice by as much as 70% compared to the original experimental design. This translates to graduation rates that are about 1.5 percentage points higher than what could be attained by policies estimated with the original Progresa experiment. This dramatic improvement comes from using the second wave to offer large experimental subsidies to small treated groups in subpopulations where the response to the transfer is potentially high (as indicated by the first wave) but still uncertain. For these subpopulations, the value to the decision maker of reducing uncertainty about the effect of the transfer is high. This second wave design therefore delivers more precise estimates of the effect of increasing the cash transfer on the subpopulations for which the value of that information is greatest.

Our approach applies to a wide range of models and objectives. While we focus on parametric models, we do not impose further restrictions on the model beyond standard regularity conditions. Parametric structural models are widely used in economics to evaluate counterfactual policies not directly observed in the data (see Petra E Todd and Kenneth I Wolpin (2006) and Attanasio, Meghir, and Santiago (2012) for examples in Progresa). We require the welfare functions to be strongly concave in the policy in a neighborhood of the optimal policy, but do not assume a specific functional form for the welfare function. This allows the decision maker to target objectives that may not be simple reduced-form functions of observed outcomes. For example, anti-poverty programs in developing countries may aim to maximize the long-run subjective utility of the recipients. Firms may seek to maximize long-run profits, mediated through the choices of forward-looking agents. A government may not be able to directly experiment with a minimum wage, but may be able to learn about labor supply elasticities through a cash transfer experiment and thereby in-

form the choice of minimum wage policy. The methods of this paper therefore enable researchers to take advantage of the “best of both worlds” described by Petra E. Todd and Kenneth I. Wolpin (2023) to leverage experimental variation and economic structure to efficiently learn about optimal policies.

1.1 *Related Literature*

This paper intersects with two complementary literatures in econometrics: experimental design and policy choice. The questions of how to design experiments, and of how to choose policies based on the results of experiments, are often treated separately in econometrics. The bandit literature, which does treat these questions jointly, does not generally achieve the asymptotic optimality results that we achieve here in our constrained, nonlinear, continuous-treatment setting.

The optimal design of experiments constitutes a vast literature in many fields. The classic approach aims to estimate the parameters of a parametric model under various optimality criteria (Silvey 2013, Pukelsheim 2006, Chaloner and Verdinelli 1995). More recent works in econometrics have focused on experimental design with the aim of efficiently estimating treatment effects with binary or discrete treatments in semiparametric settings. This literature is reviewed in S. Athey and Imbens (2017). Examples include Hahn, Hirano, and Karlan (2011), Bai (2022), Viviano (2020), Tabord-Meehan (2023), Cytrynbaum (2021), and Bai et al. (2026). The methods proposed by these papers, like the method proposed here, rely on large first-wave samples to inform the design of a second wave. When this assumption fails, such methods can have poor finite-sample properties (Cai and Rafi 2024). Asymptotic efficiency bounds for ATE estimation across experimental designs are studied in T. B. Armstrong (2022). Our setting is parametric, but we view the contribution of an asymptotically optimal experiment for policy choice in parametric models as both a novel and necessary first step towards designing experiments for policy choice in more flexible settings.

Another large literature in econometrics focuses on policy choice, given experimental or observational data. This is often done in a nonparametric or semiparametric setting (Manski 2004, Bhattacharya and Dupas 2012, Kitagawa and Tetenov 2018, Susan Athey and Wager 2021, Mbakop and Tabord-Meehan 2021, among others) or a Gaussian finite-sample environment (Stoye 2009, 2012). Counterfactual evaluation and policy choice is also a central goal of structural econometric models; we motivate

our setting with the examples of Petra E Todd and Kenneth I Wolpin (2006) and Attanasio, Meghir, and Santiago (2012), which study the Progres program that we use in our application.

The econometric policy choice literature takes the data-generating process to be outside the control of the decision maker. We complement this literature by providing guidance on how to design an experiment that will be used to choose a policy. That is, we show how to choose the most favorable data-generating process for the policy choice problem from a given class of experiments. In particular, we extend the asymptotic analysis of treatment choice rules as in Hirano and Porter (2009), Hirano and Porter (2020), and Xu (2026) to general nonlinear constrained decision problems and optimize the value of this problem across experimental designs.

Adaptive experiments are commonly used to optimize a measure of welfare, with the multi-armed bandit literature providing many algorithms for both in-sample and out-of-sample welfare maximization (Lattimore and Szepesvári (2020) and Russo et al. (2018) provide surveys, while Viviano and Rudder (2020), Kato et al. (2025), and Cesa-Bianchi, Colomboni, and Kasy (2025) provide applications to policy choice in economics). Much of the bandit literature proposes heuristic algorithms and proves rate-optimality results for expected regret. We take a different approach, related to Hirano and Porter (2025) and Adusumilli (2025a,b, 2026), in which we characterize the limiting behavior of adaptive experiments and derive designs and policies which are optimal in the limit experiment, rather than rate-optimal in finite samples. Our asymptotic representation theorem differs from prior work by allowing for continuous treatments. Our decision problem is also distinct, focusing on strongly concave welfare objectives and nonlinear constraints, as opposed to a linear objective such as treatment choice with finitely many arms.

The decision problem we study is an instance of Wald’s statistical decision theory with Bayes risk (Manski 2021). We use asymptotic approximations to reduce the dimension of the decision problem, and achieve local asymptotic Bayes optimality rather than an exact finite-sample solution (Hirano 2026). Thus, this paper contributes to the growing literature which applies the Wald framework to econometric practice (T. Armstrong, Kitagawa, and Tetenov 2025).

1.2 Outline

In Section 2, we describe the general framework and the decision problem faced by the decision maker, and find that an exact solution is infeasible. In Section 3, we describe a tractable solution method in which the difficult decision problem is replaced by a simpler Gaussian experiment with quadratic loss. In Section 4, we show the proposed method is asymptotically optimal, using a limit-of-experiments framework. In Section 5, we apply the method to the Progresca conditional cash transfer experiment.

2 ENVIRONMENT AND DECISION PROBLEM

The decision maker must choose two things: how to design the second wave of the experiment, and what policy to choose after observing the results of the experiment. The decision maker’s objective is to minimize expected regret.

2.1 Welfare and expected regret

The decision maker has a welfare function

$$W(\pi, \theta)$$

where $\pi \in \mathbb{R}^k$ is a policy which the decision maker can choose and $\theta \in \mathbb{R}^\ell$ is a finite-dimensional unknown parameter. The set of feasible policies is described by nonlinear constraints

$$\Pi = \{\pi \in \mathbb{R}^k : g(\pi) \leq 0\},$$

which may include budget constraints, fairness constraints, or other exogenous constraints on the policy, such as a nonnegativity constraint on a subsidy or price.

The welfare function is not assumed to have a specific structure beyond smoothness and local concavity assumptions specified in Section 4. The mapping from policy and parameter to welfare is context-specific, depending both on the model being used to predict the effects of counterfactual policies and the preferences of the decision maker. For example, a decision maker may be a firm choosing prices to maximize profit, a regulator setting emissions standards to maximize social welfare, or a benevolent planner choosing a tax and transfer regime. In each of these examples, $W(\pi, \theta)$

measures the welfare when policy π is chosen and consumer demand, household preferences, or the distribution of income shocks are governed by θ .¹

Since the true value of θ is not known, the decision maker evaluates policies based on their expected regret. Regret is the welfare loss from choosing a policy π and is given by

$$R(\pi, \theta) = \max_{\tilde{\pi} \in \Pi} W(\tilde{\pi}, \theta) - W(\pi, \theta)$$

where the first term is the maximum welfare achievable if θ were known. Expected regret is calculated by averaging regret across possible values of θ using a prior density $q(\theta)$, resulting in the Bayes regret objective

$$\mathbb{E}[R(\pi, \theta)] = \int R(\pi, \theta) q(\theta) d\theta.$$

Minimizing Bayes regret is equivalent to maximizing Bayes welfare because the re-centering term in regret does not affect the choice of policy. However, recentering the welfare function in this way is useful for asymptotic analysis.

While we use Bayes regret as the objective, the method proposed in this paper does not depend on the prior, and hence a user of our method does not need to be able to articulate their prior. This is because the decision maker uses the first wave to inform future decisions, and as the size of the first wave grows the influence of the prior vanishes under the conditions given in Section 4.² Our theoretical results evaluate the performance of the method from an ex-ante perspective, before any data is observed, integrating over the unknown parameter using the prior.

2.2 The data-generating process

The decision maker has access to an experimental sample. We consider a two-wave experimental design consisting of n experimental units, n_1 of which appear in the first

¹As another example case, a decision maker could have the squared error welfare function $W(\pi, \theta) = -\|\pi - \theta\|^2$, in which case the objective is to estimate θ precisely. This objective is frequently considered in the experimental design literature, often in semiparametric settings (e.g. Hahn, Hirano, and Karlan (2011)). In this case, minimizing expected regret is equivalent to minimizing the mean squared error of the estimate of θ .

²If the decision maker has prior information that they do not wish to ignore even with large samples, this can be incorporated by specifying the prior to be over the local alternatives described in Section 4. In this case, the local prior must be Gaussian for the method to remain tractable.

wave and n_2 of which appear in the second wave. We denote the units in each wave by $\mathcal{I}_{1,n} = \{1, \dots, n_1\}$ and $\mathcal{I}_{2,n} = \{n_1 + 1, \dots, n\}$ respectively. Each unit is associated with a vector of covariates x_i . Our analysis is conditional on covariates, and we treat them as fixed throughout.³ During the experiment, each unit is assigned a treatment z_i , after which an outcome y_i is observed.

In the first wave, the treatment z_i is assigned according to a fixed experimental design which is not chosen by the decision maker. Treatment may be discrete or continuous, and potentially multidimensional. The distribution of treatment conditional on covariates is determined by the density

$$p_{z|x}(z_i \mid x_i; \delta_1), \quad i \in \mathcal{I}_{1,n}$$

where δ_1 is a fixed parameter describing the design of the first wave. The dominating measure for the density $p_{z|x}$ may be Lebesgue measure or another measure. For example, a binary treatment may be assigned by the mechanism $z_i = \mathbb{1}[\delta'_1 x_i \geq \nu_i]$ where ν_i is a uniform randomization device, while a continuous treatment may be assigned by $z_i = \delta'_1 x_i + \nu_i$ where ν_i is a Gaussian randomization device. We leave the dominating measure unspecified, allowing for atoms in the distribution of z_i .

After treatment is assigned in the first wave, the outcome y_i is observed for these units. The outcome may not directly measure welfare, but observations of y_i are informative about θ , which determines the welfare effects of policies. Outcomes are generated according to the density

$$p_{y|z,x}(y_i \mid z_i, x_i; \theta), \quad i \in \mathcal{I}_{1,n}$$

where $\theta \in \Theta$ is an unknown finite-dimensional parameter and Θ is an open subset of $\mathbb{R}^\ell$. Even though we restrict to parametric models, our framework can be used with nonlinear models capturing rich economic behavior. For example, in Section 5 we use a dynamic school choice model with forward-looking households. In principle, this model could be extended to include fertility decisions and general equilibrium effects, as in Petra E Todd and Kenneth I Wolpin (2006) and Attanasio, Meghir, and Santiago (2012).

While we refer to the observations $i \in \mathcal{I}_{1,n}$ as the first wave of the experiment, this

³Alternatively, we could assume a parametric model $p_x(x_i; \theta)$ for the covariates as in Hirano and Porter (2025).

data can take many forms in practice. It may arise from an experiment conducted by the decision maker or by another entity. It may be an observational dataset if the decision maker believes that θ is identified from the observational data. It can even be replaced by an asymptotically informative prior as in Adusumilli (2025b). The first wave represents existing data available to the decision maker that may be informative for policy decisions, but does not reveal the optimal policy with perfect certainty.

In the second wave, the decision maker may choose the design δ_2 governing the conditional distribution of treatment for the n_2 units in the second wave.⁴ The design δ_2 for the second wave can specify how the probability of treatment depends on covariates or how the value of a continuous treatment (such as a price) depends on covariates. It is restricted to lie in a compact set

$$\delta_2 \in \Delta$$

which describes constraints on the experimental design, such as budget constraints, ethical constraints, etc. Once δ_2 has been decided, the treatment for units in the second wave is drawn according to

$$p_{z|x}(z_i \mid x_i; \delta_2), \quad i \in \mathcal{I}_{2,n}.$$

As in the first wave, δ_2 may parametrize a discrete or continuous distribution of z_i given x_i .

Once the decision maker assigns treatment, outcomes for the second wave are observed. These outcomes are drawn according to the same conditional density as in the first wave,

$$p_{y|z,x}(y_i \mid z_i, x_i; \theta), \quad i \in \mathcal{I}_{2,n},$$

so both experimental waves can be combined to learn about θ .

⁴The size of the second wave may be endogenized by introducing a sampling variable $\tilde{z}_i$ which is Bernoulli with probability determined by δ_2 . If the unit is sampled, y_i is observed; otherwise it is missing. In this case, for the theoretical results that follow we would need to ensure that the sampling probability be bounded away from zero. If $z_i = \tilde{z}_i$ is the only variable assigned by the decision maker, the experiment can be viewed simply as a survey in which the outcome y_i is collected from a subset of the units, with sampling probabilities described by δ_2 . Thus, the method proposed in this paper can also be used to design surveys which inform policy choice.

We restrict attention to experimental designs in which treatment is assigned independently across units within a wave, which simplifies the asymptotic analysis of Section 4. A large literature notes the potential precision gains from designs with dependent treatment assignments such as complete randomization, matched pairs, or biased coins (see the literature review for references). We defer the analysis of such designs in our setting to future work.

2.3 Policy choice and the dynamic decision problem

After observing the outcomes of both waves of the experiment, the decision maker chooses a policy π to minimize posterior expected regret. Letting the first-wave and second-wave random data be denoted by

$$D_{1,n} = \{y_i, z_i\}_{i \in \mathcal{I}_{1,n}}, \quad D_{2,n} = \{y_i, z_i\}_{i \in \mathcal{I}_{2,n}},$$

the value of an experimental dataset is given by

$$V_n(D_{1,n}, D_{2,n}) = \min_{\pi} \mathbb{E}[R(\pi, \theta) \mid D_{1,n}, D_{2,n}] \quad \text{s.t.} \quad g(\pi) \leq 0. \quad (1a)$$

By backwards induction, the optimal design of the second wave is the solution to

$$\min_{\delta_2} \mathbb{E}_{\delta_2} [V_n(D_{1,n}, D_{2,n}) \mid D_{1,n}] \quad \text{s.t.} \quad \delta_2 \in \Delta \quad (1b)$$

where $\mathbb{E}_{\delta_2}$ denotes the expectation with respect to the law of motion for the second wave of the experiment when the design δ_2 is chosen,

$$y_i, z_i \sim p_{y|z,x}(y_i \mid z_i, x_i; \theta) p_{z|x}(z_i \mid x_i; \delta_2), \quad i \in \mathcal{I}_{2,n} \quad (1c)$$

in which treatment z_i is first assigned according to δ_2 and then the outcome is realized according to the unknown parameter θ .

The dynamic program (1) characterizes the optimal design for the second wave, but it is infeasible to solve in practice. One could imagine trying to solve it by standard dynamic programming methods, but since π is a function of all experimental data, the state space of V_n is the set of possible datasets that the second wave of the experiment could generate. This is extremely high-dimensional even for moderate-sized

experiments and low-dimensional variables⁵. In the example considered in Section 5, y_i is binary, $z_i \in \mathbb{R}^7$, and $n_2 = 1000$, resulting in a state space of $\{0, 1\}^{1000} \times \mathbb{R}^{7000}$.

This problem is a general feature of dynamic experiments, and features prominently in multi-armed bandit literature. More broadly, it is a feature of statistical decision problems where a decision maker must specify a decision rule which may depend on realizations of the data (Manski 2021). For this reason, much of the literature on adaptive experimentation has proposed heuristic algorithms which are approximate solutions to the finite-sample problem (Lattimore and Szepesvári 2020).

We propose a general-purpose solution method for this problem motivated by asymptotic approximations. In contrast to much of the existing literature on adaptive experimentation, our method is not specific to a particular welfare function or model, such as the multi-armed bandit setting, but accommodates general constrained nonlinear decision problems. Moreover, our method provides guarantees not just on the rate of convergence but also on the optimality of the expected regret in the limit experiment, which allows us to distinguish between designs which achieve the same convergence rate for expected regret.

3 SOLUTION METHOD

Since the finite-sample problem is intractable, we propose an approximation to this problem that is low-dimensional and straightforward to solve. This approximation has two components: first, replacing the nonlinear regret with a quadratic function, and second, replacing the data-generating process with a Gaussian observation. Following the logic of backward induction, we present these approximations in reverse order: the quadratic approximation reduces the dimension of the terminal value function, and the Gaussian approximation simplifies the law of motion for the state of the dynamic program.

This section presents the method as it is implemented in practice. Formal justification of the approximations and the asymptotic optimality of the method are given in Section 4.

⁵Alternatively, one could use the posterior on θ as the state, but if the likelihood and prior are not conjugate the state space will be the set of probability distributions on $\mathbb{R}^\ell$.

3.1 Quadratic approximation of expected regret

We replace the nonlinear expected regret loss function with a quadratic approximation. Under this approximation, the optimal policy depends on the unknown parameter θ only through a particular linear functional, which is approximately equal to the marginal effect of changing the policy. This implies that the state of the dynamic program can be reduced to the posterior mean of this marginal effect.

Our quadratic approximation is based on a second-order Taylor expansion of the regret function around a reference point. Suppose that the true parameter θ is in a neighborhood of some θ_0 . This neighborhood will be formalized in Section 4. Let the optimal policy under θ_0 be

$$\pi_0 = \arg \min_{\pi} R(\pi, \theta_0) \quad \text{s.t.} \quad g(\pi) \leq 0. \quad (2)$$

The centering term in regret makes $R(\pi, \theta)$ potentially nondifferentiable in θ . Our method instead uses derivatives of the welfare function $W(\pi, \theta)$, which is smooth in θ , and ignores the policy-irrelevant centering term in regret. Define

$$D = -\nabla_{\pi} W(\pi_0, \theta_0), \quad B = -\nabla_{\pi\theta}^2 W(\pi_0, \theta_0), \quad H = -\nabla_{\pi\pi}^2 W(\pi_0, \theta_0).$$

Since the choice of policy is subject to constraints, it will not generally be true that $D = 0$, so this first-order term must be included in the approximation. The quadratic function we minimize is then given by

$$\tilde{R}(\pi, \beta) = (\pi - \pi_0)' D + (\pi - \pi_0)' (\beta - \beta_0) + \frac{1}{2} (\pi - \pi_0)' H (\pi - \pi_0)$$

where

$$\beta = B\theta, \quad \beta_0 = B\theta_0.$$

The approximation $\tilde{R}$ depends on θ only through the linear functional β . The marginal effect of changing the policy in a neighborhood of (π_0, θ_0) is

$$\nabla_{\pi} R(\pi_0, \theta) \approx D + (\beta - \beta_0)$$

and so the parameter β gives an approximation to the marginal effect of the policy

in this neighborhood. When choosing a policy, estimates of the parameters of the model are only relevant insofar as they are informative about the effects of counterfactual policies. For policies close to π_0 , the marginal effect of changing the policy characterizes these counterfactuals. Thus, when choosing a policy, the decision maker need only consider the posterior on this marginal effect. In the Progresa example, this marginal effect is the increase in graduation rates resulting from giving an extra peso to boys or girls in primary or secondary school.

The key benefit of this approximation is that $\tilde{R}$ is linear in β . Denoting the posterior mean of β after both waves of the experiment by $\hat{\beta}_{2,n}$, we have

$$\mathbb{E} \left[\tilde{R}(\pi, \beta) \mid D_{1,n}, D_{2,n} \right] = \tilde{R}(\pi, \hat{\beta}_{2,n})$$

and so when choosing a policy to minimize the expectation of $\tilde{R}(\pi, \beta)$, the decision maker only needs to consider the posterior mean of β .

The value function under this quadratic approximation is given by

$$\tilde{V}_n(\hat{\beta}_{2,n}) = \min_{\pi} \tilde{R}(\pi, \hat{\beta}_{2,n}) \quad \text{s.t.} \quad g(\pi) \leq 0.$$

This value function is much lower-dimensional than the original value function V_n . The state is simply $\hat{\beta}_{2,n}$, rather than the full experimental data $D_{1,n}$ and $D_{2,n}$. In contrast to V_n , the state of $\tilde{V}_n$ does not grow with the sample size. In the Progresa application in Section 5, the policy is four-dimensional—consisting of a subsidy amount for boys and girls in primary and secondary school—and so the marginal effect β is also four-dimensional, which makes computing $\tilde{V}_n$ straightforward in practice. However, the law of motion for $\hat{\beta}_{2,n}$ is still complicated, motivating our next approximation.

3.2 Gaussian approximation of estimates

While the quadratic approximation reduces the dimensionality of the state space, choosing an experimental design still requires minimizing the expected value function over the distribution of the data from the second wave of the experiment. This is a high-dimensional law of motion for $\hat{\beta}_{2,n}$. Our second approximation is to replace the data-generating process with a Gaussian observation. Rather than observing the full data from each wave, we assume that the decision maker only observes a Gaussian

estimate from each wave, with precision determined by δ_2 . This allows the decision maker to reduce the problem of how to assign treatments to the problem of finding the optimal posterior variance-covariance matrix for $\hat{\beta}_{2,n}$.

We approximate the prior distribution and experimental data with a flat prior and Gaussian observations, respectively. After the first wave, suppose the decision maker only observes a Gaussian estimate $\hat{\theta}_{1,n}$ of θ , which satisfies

$$\hat{\theta}_{1,n} \stackrel{a}{\sim} N\left(\theta, \frac{1}{n_1} J_{1,n}^{-1}\right)$$

where

$$J_{1,n} = \frac{1}{n_1} \sum_{i \in \mathcal{I}_{1,n}} \mathbb{E} \left[\nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \theta_0) \nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \theta_0)' \right]$$

is the Fisher information matrix at θ_0 for the first wave of the experiment. With a flat prior, the posterior distribution of θ given $\hat{\theta}_{1,n}$ is Gaussian, with mean $\hat{\theta}_{1,n}$ and covariance matrix $\frac{1}{n_1} J_{1,n}^{-1}$. After choosing δ_2 and running the second wave of the experiment, suppose the decision maker observes a second Gaussian estimate $\hat{\theta}_{2,n}$ from the second wave, which satisfies

$$\hat{\theta}_{2,n} \stackrel{a}{\sim} N\left(\theta, \frac{1}{n_2} J_{2,n}(\delta_2)^{-1}\right)$$

where

$$J_{2,n}(\delta_2) = \frac{1}{n_2} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}_{\delta_2} \left[\nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \theta_0) \nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \theta_0)' \right]$$

is the Fisher information matrix when the design δ_2 is chosen, and $\mathbb{E}_{\delta_2}$ indicates that z_i is assigned according to the design δ_2 . By varying the experimental design δ_2 , the decision maker varies the variance-covariance matrix of the resulting estimate $\hat{\theta}_{2,n}$.

The information matrix $J_{2,n}(\delta_2)$ affects the welfare of the policy chosen at the end of the experiment by determining the information available to the decision maker when choosing the policy. This is captured by the posterior distribution of β conditional on the Gaussian estimates $\hat{\theta}_{1,n}$ and $\hat{\theta}_{2,n}$. This posterior is also Gaussian and is given by the standard Bayesian updating formula for Gaussian distributions. An implication of this is that our method will not depend on the choice of prior distribu-

tion. This property may be desirable if the decision maker has difficulty articulating their prior, but it may be undesirable if the decision maker has a strong prior belief that is not reflected by a flat or weakly informative prior. The method here can easily extend to use Gaussian approximations to informative priors, but we focus on the flat prior case for simplicity.

Designs which are more informative about particular parameters will lead to posteriors which are more informative about those parameters. Whether a particular δ_2 is optimal in terms of its information depends on the objective and constraints of the decision maker. In particular, our quadratic approximation implies that β is a sufficient statistic for the choice of policy. However, due to the constraints on the policy choice and the local curvature of welfare, the decision maker may not care about all components of β equally. If some margins described by β are infeasible, or if the first wave of the experiment suggests that some margins would almost certainly decrease welfare, then the decision maker may not want to spend their experimental budget to learn about those margins.

The posterior mean $\hat{\beta}_{2,n}$ under the Gaussian approximation is given by

$$\hat{\beta}_{2,n} = B (n_1 J_{1,n} + n_2 J_{2,n}(\delta_2))^{-1} \left(n_1 J_{1,n} \hat{\theta}_{1,n} + n_2 J_{2,n}(\delta_2) \hat{\theta}_{2,n} \right).$$

Equivalently, since we are using a flat prior approximation, $\hat{\beta}_{2,n}$ is the pooled maximum likelihood estimate of β from the two waves of the experiment. This data-generating process is much simpler than the original data-generating process for the dynamic program (1), which involves the full data from both waves of the experiment. As a result, minimizing the expectation of $\tilde{V}_n$ over the Gaussian approximation to the distribution of $\hat{\beta}_{2,n}$ is much simpler than minimizing over the distribution of the full data. In particular, Gaussian approximation to the law of motion for $\hat{\beta}_{2,n}$ given the observation $\hat{\theta}_{1,n}$ from the first wave of the experiment is given by

$$\hat{\beta}_{2,n} \sim N \left(\hat{\beta}_{1,n}, B \left(\frac{1}{n_1} J_{1,n}^{-1} - (n_1 J_{1,n} + n_2 J_{2,n}(\delta_2))^{-1} \right) B' \right) \mid \hat{\theta}_{1,n}$$

which is much easier to integrate over than the law of motion for $D_{2,n}$ given $D_{1,n}$.

Finally, we note that our method still works if δ_2 is chosen such that $J_{2,n}(\delta_2)$ is singular, as long as the resulting posterior information $n_1 J_{1,n} + n_2 J_{2,n}(\delta_2)$ is non-singular. In this case, the MLE $\hat{\theta}_{2,n}$ may not be well-defined, but the pooled MLE

from both waves, and the posterior mean $\hat{\beta}_{2,n}$, will still be well-defined. In particular, the posterior distribution and law of motion given above do not require $J_{2,n}(\delta_2)$ to be non-singular. In fact, choosing a singular $J_{2,n}(\delta_2)$ may be optimal in some cases. This is because not all components of θ may be relevant for the policy choice, and so a decision maker may want to choose a design that is informative about β but provides no information about other components of θ .

3.3 Feasible analog and value function

The approximation described above is constructed around a reference point (π_0, θ_0) which is unknown to the decision maker. To make this method feasible, we replace the unknown reference point with estimates from the first wave of the experiment. We use these estimates to construct empirical analogues of the Gaussian and quadratic approximations described above.

After collecting the first wave of data, the decision maker constructs the maximum likelihood estimator $\hat{\theta}_{1,n}$, which is approximately normally distributed. The decision maker then constructs an estimate of the optimal policy from the first wave, given by

$$\hat{\pi}_{1,n} = \arg \min_{\pi} R(\pi, \hat{\theta}_{1,n}) \quad \text{s.t.} \quad g(\pi) \leq 0.$$

The first-wave estimates $(\hat{\pi}_{1,n}, \hat{\theta}_{1,n})$ are then used as plug-in estimates for the unknown reference point (π_0, θ_0) . The quadratic approximation to regret is estimated by

$$\hat{R}_n(\pi, \beta) = (\pi - \hat{\pi}_{1,n})' \hat{D}_n + (\pi - \hat{\pi}_{1,n})' (\beta - \hat{\beta}_{1,n}) + \frac{1}{2} (\pi - \hat{\pi}_{1,n})' \hat{H}_n (\pi - \hat{\pi}_{1,n})$$

where $\hat{D}_n$, $\hat{\beta}_{1,n} = \hat{B}_n \hat{\theta}_{1,n}$, and $\hat{H}_n$ are all given by plug-in estimates. Likewise, the Gaussian approximation is estimated by replacing the unknown Fisher information matrices $J_{1,n}$ and $J_{2,n}(\delta_2)$ with plug-in estimates. These are given by

$$\begin{aligned} \hat{J}_{1,n} &= \frac{1}{n_1} \sum_{i \in \mathcal{I}_{1,n}} \nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \hat{\theta}_{1,n}) \nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \hat{\theta}_{1,n})' \\ \hat{J}_{2,n}(\delta_2) &= \frac{1}{n_2} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}_{\delta_2} \left[\nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \hat{\theta}_{1,n}) \nabla_{\theta} \log p_{y|z,x}(y_i | z_i, x_i; \hat{\theta}_{1,n})' \right] \end{aligned}$$

where the expectation in the second line is taken over the distribution of z_i when the design is δ_2 and the distribution of y_i when the parameter is $\hat{\theta}_{1,n}$, both of which are

known to the decision maker after the first wave of the experiment.

Thus, the feasible approximation to the value function which the decision maker can solve in practice is

$$\hat{V}_n(\hat{\beta}_{2,n}) = \min_{\pi} \hat{R}_n(\pi, \hat{\beta}_{2,n}) \quad \text{s.t.} \quad g(\pi) \leq 0 \quad (3a)$$

after which the decision maker chooses the design of the second wave to minimize the expected value function:

$$\min_{\delta_2} \mathbb{E} \left[\hat{V}_n(\hat{\beta}_{2,n}) \mid \hat{\theta}_{1,n} \right] \quad \text{s.t.} \quad \delta_2 \in \Delta \quad (3b)$$

where the expectation integrates over the second-wave distribution of $\hat{\beta}_{2,n}$ using a plug-in estimate of the Gaussian Bayesian updating formula for the law of motion:

$$\hat{\beta}_{2,n} \sim N \left(\hat{\beta}_{1,n}, \hat{B}_n \left(\frac{1}{n_1} \hat{J}_{1,n}^{-1} - (n_1 \hat{J}_{1,n} + n_2 \hat{J}_{2,n}(\delta_2))^{-1} \right) \hat{B}'_n \right) \mid \hat{\theta}_{1,n}. \quad (3c)$$

We denote the solution to this problem by $\hat{\delta}_{2,n}$.

Solving the value function $\hat{V}_n$ in practice is straightforward, whereas solving V_n is generally intractable. While algorithmic details are deferred to Appendix E, the main steps are (i) evaluate $\hat{V}_n(\beta)$ for many values of β , (ii) interpolate to construct a smooth surrogate for $\hat{V}_n$, (iii) compute $\mathbb{E}_{\delta}[\hat{V}_n(\hat{\beta}_{2,n}) \mid \hat{\theta}_{1,n}]$ as a function of δ by Monte Carlo integration (in our application, this is differentiable in δ), and (iv) optimize over δ_2 using a nonlinear optimization routine. Solving $\hat{V}_n$ is especially tractable when the constraints $g(\pi) \leq 0$ are linear, as is the case in the Progres application in Section 5. Then $\hat{V}_n(\hat{\beta})$ is a quadratic program for any value of $\hat{\beta}$, making construction of $\hat{V}_n$ very fast.

4 ASYMPTOTIC OPTIMALITY

The solution method presented in the previous section is asymptotically optimal. We establish this by showing that the approximations made in the previous section have negligible error as the sizes of the two waves of the experiment grow large. These results imply that the design $\hat{\delta}_{2,n}$ that comes from solving the approximated value function leads to the lowest possible regret asymptotically.

4.1 Local asymptotic framework

The Gaussian and quadratic approximations are justified in an asymptotic decision environment called the limit experiment. The limit experiment describes the possible asymptotic behaviors of any design and policy in the finite-sample environment. The benefit of working in the limit experiment is that it is much simpler than the finite-sample problem, allowing us to characterize the best possible design and policy in the limit.

The limit experiment is constructed in a local asymptotic framework. Local asymptotics are commonly used to study optimality of estimators, tests, and the consequences of model misspecification, among other econometric and statistical problems (Le Cam 1972, Van der Vaart 2000).

In local asymptotics, we analyze the performance of our method under alternative parameter values that are difficult to distinguish from each other even in large samples. In particular, we model

$$\theta = \theta_0 + h/\sqrt{n}$$

for some “local parameter” h and reference parameter θ_0 . The $1/\sqrt{n}$ scaling means that the difference between θ and θ_0 is of the same order of magnitude as the standard error of typical estimators of θ . Likewise, we will consider local alternatives to a reference policy π_0 , where

$$\pi = \pi_0 + c/\sqrt{n}$$

so that the decision maker considers policies in a neighborhood of the reference policy π_0 of the same order of magnitude as sampling uncertainty. Regret can be written in terms of local parameters as

$$R_n(c, h) = nR(\pi_0 + c/\sqrt{n}, \theta_0 + h/\sqrt{n})$$

where the n scaling ensures a nontrivial limit as $n \rightarrow \infty$.

Like all asymptotics, local asymptotics are not meant to reflect a data-generating process that is literally changing as the sample size grows, but rather is intended to capture the finite-sample property that θ is difficult to distinguish from θ_0 at the current sample size in two senses which are material for our approximation. First,

θ_0 and π_0 are used to define the quadratic approximation to regret. Therefore, we need welfare to be well-approximated by a quadratic function for values of θ and π that are plausible after the first wave of the experiment. This can be directly evaluated by sampling from the posterior distribution of θ and the implied posterior distribution of π after the first wave of the experiment, and seeing whether welfare is approximately quadratic over the support of these distributions. Second, θ_0 is used to define the Gaussian law of motion for $\hat{\beta}_{2,n}$. Thus, we need estimates of $\hat{\beta}_{2,n}$ to be approximately Gaussian with a covariance matrix that is well-approximated by $J_2(\cdot)$ for values of θ that are plausible after the first wave of the experiment. This can also be informally assessed by checking the posterior distribution of β after the first wave of the experiment.

Local asymptotics are also essential for our asymptotic framework to capture finite-sample uncertainty about which constraints are binding. The set of θ for which a particular constraint holds with weak complementary slackness generally has Lebesgue measure zero. This means that for almost every θ , the decision problem behaves asymptotically as if the set of binding constraints is known. In these cases, the decision problem in the limit experiment reduces to minimizing a quadratic function over a linear subspace, and hence is equivalent to point estimation of a linear functional of θ . In contrast, our approach of localizing around a reference parameter θ_0 where weak complementary slackness holds preserves nontrivial constraints in the limit experiment. Thus, we can focus on instances of the problem in which even asymptotically, the decision maker does not know whether the optimal policy lies on the boundary of the feasible set. This uncertainty is essential to understanding how policy constraints affect the value of different experimental designs.

This localization has an intuitive interpretation in the cash-transfer example of Section 5. The decision maker is most likely to make a mistake when it is difficult to detect differences in the marginal effects of the cash transfer across groups. When this is the case, noisy experimental data may lead the decision maker to allocate extra money to boys even when girls would benefit more. However, it is reasonable to expect that the effect of the cash transfer is positive, so the budget constraint is likely to bind and the optimal policy is likely to be on the boundary of the feasible set. In this scenario, we take θ_0 to be a point where the marginal effects of the cash transfer are positive and equal across groups, and π_0 is a vertex of the feasible set. Then at π_0 , the cash is allocated to only one of the four groups, but local perturbations $h \neq 0$

may cause the optimal policy c to move away from the vertex and allocate some cash to the other groups. Our asymptotic framework does not require that θ_0 be specified, but it is intended to capture this statistically “hard” instance of the decision problem. This is analogous to localizing around an average treatment effect of zero in a binary treatment choice problem (Hirano and Porter 2009).

4.2 Limit experiment

The bound we establish on the limiting regret of any sequence of designs is given by the value of an optimization problem in the limit experiment. This limiting decision problem is a quadratic program with linear constraints.

The constraints in the limit experiment are given by linear approximations to the constraints in the finite-sample problem. Define λ_0 to be the unique Lagrange multiplier (guaranteed to exist under the assumptions given below) satisfying

$$D + \lambda_0' \nabla g(\pi_0) = 0 \quad \lambda_{0j} \geq 0 \quad \lambda_{0j} g_j(\pi_0) = 0$$

for $j = 1, \dots, J$. Then define the Hessian of the Lagrangian at (π_0, λ_0) as

$$\bar{H} = H + \lambda_0' \nabla^2 g(\pi_0).$$

Let the set of indices j for which strict complementary slackness holds at π_0 be

$$\mathcal{J}_s = \{j : g_j(\pi_0) = 0, \lambda_{0j} > 0\}$$

Likewise, let the set of indices for which weak complementary slackness holds at π_0 be

$$\mathcal{J}_w = \{j : g_j(\pi_0) = \lambda_{0j} = 0\}.$$

Taking the gradients of the constraints at π_0 , we obtain a convex cone which is the feasible set in the limit experiment,

$$\mathcal{G} = \{c : \nabla g_j(\pi_0)' c = 0, j \in \mathcal{J}_s, \nabla g_j(\pi_0)' c \leq 0, j \in \mathcal{J}_w, \}.$$

The quadratic objective arises from the quadratic approximation to regret, but

also incorporates the curvature of the constraints through the Hessian of the Lagrangian. We first define the quadratic objective

$$r(c, b) = c'b + \frac{1}{2}c'\bar{H}c$$

where $b = Bh$ is the local alternative for β . This quadratic objective, like $\tilde{R}$ in Section 3, omits the policy-irrelevant centering term in regret, which simplifies the dynamic program in the limit experiment. The asymptotic lower bound, however, is given by the regret in the limit experiment, which is

$$R_\infty(c, b) = r(c, b) - \min_{c \in \mathcal{G}} r(c, b).$$

Having described the objective and constraints in the limit experiment, we now describe the information structure in the limit experiment. We begin with the prior in the limit experiment. Fix $K > 0$ and define the localized finite-sample prior as

$$q_{n,K}(h) \propto q(\theta_0 + h/\sqrt{n})\mathbb{1}\{\|h\| \leq K\}$$

normalized to integrate to one. The corresponding truncated density in the limit experiment is

$$q_{\infty,K}(h) \propto \mathbb{1}\{\|h\| \leq K\}$$

which is uniform on the ball of radius K . This localization of the prior serves two purposes. First, it ensures that the limit of expected regret is finite, and hence the asymptotic lower bound is well-defined. Second, as discussed in Section 4.1, it reflects the finite-sample property that some constraints may be on the margin of binding or nonbinding, since this prior asymptotically places mass on parameters which are local to θ_0 , a point of possible weak complementary slackness. In the results that follow, we will show our method, which is motivated by a flat-prior approximation, is asymptotically optimal as K grows large, and hence as our local prior becomes weakly informative.

The observed data in the limit experiment comes from a two-wave adaptive experiment with Gaussian signals. Let ρ denote the asymptotic fraction of the sample in the second wave of the experiment. In the first wave of the limit experiment, the

decision maker observes a random variable

$$A_1 \sim N((1 - \rho)J_1 h, (1 - \rho)J_1).$$

Note that when J_1 is nonsingular, observing A_1 is equivalent to observing a Gaussian signal $\hat{h}_1 = (1 - \rho)^{-1}J_1^{-1}A_1 \sim N(h, (1 - \rho)^{-1}J_1^{-1})$, which is the limiting distribution of the first-wave maximum likelihood estimator. After observing A_1 , the decision maker chooses a design δ_2 and observes

$$A_2 \sim N(\rho J_2(\delta_2)h, \rho J_2(\delta_2)).$$

As with the first wave, when $J_2(\delta_2)$ is nonsingular, observing A_2 is equivalent to observing a Gaussian signal $\hat{h}_2 = \rho^{-1}J_2(\delta_2)^{-1}A_2 \sim N(h, \rho^{-1}J_2(\delta_2)^{-1})$. However, the possibility of singular $J_2(\delta_2)$ is important in an adaptive experiment, since it may be optimal to choose a design δ_2 which is not informative about some directions in h , since only b is relevant for the policy choice.

After observing A_1 and A_2 , the decision maker chooses a policy c to minimize the posterior expected regret. Let $\hat{b}_{2,K}$ be the posterior mean of b conditional on A_1 and A_2 under the prior $q_{\infty,K}$. The value function in the limit experiment (omitting the centering term) is

$$V(\hat{b}_{2,K}) = \min_c r(c, \hat{b}_{2,K}) \quad \text{s.t.} \quad c \in \mathcal{G}$$

and the optimal design in the limit experiment is given by

$$\min_{\delta_2} \mathbb{E}_{\delta_2,K}[V(\hat{b}_{2,K}) \mid A_1] \quad \text{s.t.} \quad \delta_2 \in \Delta.$$

We denote the optimal design in the limit experiment by $\delta_{2,K}^*$ and the optimal policy by c_K^* .

We evaluate the asymptotic performance of a sequence of designs and policies by their Bayes regret under the truncated prior $q_{n,K}$. We consider experimental designs which are described by a fixed design rule ϕ_n so that $\delta_{2,n} = \phi_n(D_{1,n}, U)$ where U is auxiliary randomness independent of the data. The ex-ante Bayes regret under $q_{n,K}$

is given by

$$\mathcal{R}_{n,K}(\delta_{2,n}, c_n) = \int \mathbb{E}_h^{\phi_n} [R_n(c_n, h)] q_{n,K}(h) dh$$

where $\mathbb{E}_h^{\phi_n}$ denotes the expectation over the data when the local parameter is h and the design rule is ϕ_n . When $c_n = c_{n,K}^*$ is the Bayes optimal policy for $\delta_{2,n}$, we abbreviate this to $\mathcal{R}_{n,K}^*(\delta_{2,n})$, and when $\delta_{2,n} = \delta_{2,n,K}^*$ is the Bayes optimal design, we abbreviate this to $\mathcal{R}_{n,K}^{**}$. The fixed- K regret of a design δ_2 and policy c in the limit experiment is

$$\mathcal{R}_{\infty,K}(\delta_2, c) = \int \mathbb{E}_h^\phi [R_\infty(c, Bh)] q_{\infty,K}(h) dh$$

where ϕ is the limit experiment design rule governing $\delta_2 = \phi(A_1, U)$. The quantities $\mathcal{R}_{\infty,K}^*(\delta_2)$ and $\mathcal{R}_{\infty,K}^{**}$ are defined analogously.

4.3 Asymptotic representation theorem

Our first result is an asymptotic representation theorem which justifies the Gaussian approximation to the information environment. It states that the statistical behavior of any sequence of designs and policies in the finite-sample experiment is asymptotically approximated by a design and policy in the Gaussian limit experiment.

To apply asymptotic approximations to both waves of the experiment, we require that both waves of the experiment grow large.

Assumption 1: *The sizes of the waves satisfy $\lim_{n \rightarrow \infty} n_2/n = \rho$ and $\lim_{n \rightarrow \infty} n_1/n = 1 - \rho$ for some $\rho \in (0, 1)$.*

This assumption implies that the first wave cannot be an uninformative pilot that is vanishingly small compared to the second wave—it must be large enough to yield a $\sqrt{n}$ -consistent estimate of θ .

A key (and standard) condition for local asymptotic analysis is that the model is smooth in θ in the sense of differentiability in quadratic mean (DQM), allowing for local approximations to the likelihood.

Assumption 2: *There exists a function $\psi(y \mid z, x)$, called the score of $p_{y|z,x}$ at θ_0 ,*

such that for all h ,

$$\sup_{z,x} \int \left[\sqrt{p_{y|z,x}(y | z, x; \theta_0 + h)} - \sqrt{p_{y|z,x}(y | z, x; \theta_0)} - \frac{1}{2} h' \psi(y | z, x) \sqrt{p_{y|z,x}(y | z, x; \theta_0)} \right]^2 \mu(dy) = o(\|h\|^2)$$

as $h \rightarrow 0$, for some measure μ which dominates $p_{y|z,x}$ for all (z, x) .

At a high level, this means the square root of the density is differentiable in θ at θ_0 . This condition allows us to use the score to linearly approximate the likelihood at θ near θ_0 . Typically, the score is the gradient of the log-likelihood of the model (see Lemma 7.6 of Van der Vaart (2000) for sufficient conditions for this formulation).

To show that the Gaussian approximation holds uniformly across the set of designs, we need some smoothness of the data-generating process as the decision maker varies the design. This smoothness is not imposed by the conditional distributions $p_{z|x}(\cdot | x; \delta)$ and $p_{y|z,x}(\cdot | z, x; \theta)$, and so we link these conditional distributions together through a potential outcomes model.

Assumption 3: *There exist functions $z(x, \nu; \delta)$ and $y(z, x, \epsilon; \theta)$ such that*

$$z_i = z(x_i, \nu_i; \delta), \quad y_i = y(z_i, x_i, \epsilon_i; \theta)$$

where ν_i and ϵ_i are independent of each other and independent across i .

The structural error ϵ_i captures the unobserved determinants of the outcome y_i , while the randomization device ν_i is generated by the decision maker to determine the treatment assignment z_i . The potential outcome model provides a link across the possible experiments the decision maker may choose, allowing us to analyze the entire decision environment. In particular, we can define the score of an observation i as a function of the design δ as

$$\psi_i(\delta) = \psi \left(y \left(z(x_i, \nu_i; \delta), x_i, \epsilon_i; \theta \right), z(x_i, \nu_i; \delta), x_i \right).$$

We also define the score process and log-likelihood ratio in the second wave as

$$A_{2,n}(\cdot) = \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} \psi_i(\cdot), \quad \Lambda_{2,n}^h(\cdot) = \sum_{i \in \mathcal{I}_{2,n}} \log \frac{p_{y|z,x}(y_i(\cdot) | z_i(\cdot), x_i; \theta_0 + h/\sqrt{n})}{p_{y|z,x}(y_i(\cdot) | z_i(\cdot), x_i; \theta_0)}.$$

The sample score $A_{1,n}$ and log-likelihood ratio $\Lambda_{1,n}^h$ in the first wave are defined analogously, but with the fixed design δ_1 .

The specific smoothness required for uniform approximations of the decision environment is stochastic equicontinuity of the log-likelihood ratio and the score as processes in Δ .

Assumption 4: Δ is a compact subset of Euclidean space. $A_{2,n}(\cdot)$ is stochastically equicontinuous as a process in Δ under θ_0 . For any h , $\Lambda_{2,n}^h(\cdot)$ is stochastically equicontinuous as a process in Δ under θ_0 .

Stochastic equicontinuity is a high-level condition that ensures that the score and log-likelihood have relatively smooth sample paths for large enough n . Many lower-level sufficient conditions are available in the literature (e.g. Andrews (1994), Van der Vaart and Wellner (2013)). While Assumption 4 is imposed only under θ_0 , we show in Appendix A that the laws of the adaptive experiments under local alternatives are contiguous, so that approximations that vanish in probability under θ_0 also vanish in probability under local alternatives.

Assumption 4 is a joint restriction on both the treatment assignment mechanism and the outcome model. Both the treatment assignment and the model must be smooth enough that small changes in δ lead to small changes in the data and associated likelihood. In Appendix D we give lower-level sufficient conditions which are natural for our setting, and verify that they are satisfied in the Progres example in Section 5.

Finally, we need the sample information matrices to have well-defined limits.

Assumption 5: There exists a matrix J_1 and a covariance kernel $J_2(\cdot, \cdot)$ such that for any $d_1, d_2 \in \Delta$,

$$\begin{aligned} \frac{1}{n_1} \sum_{i \in \mathcal{I}_{1,n}} \mathbb{E}_{\theta_0} [\psi_i(\delta_1) \psi_i(\delta_1)'] &\rightarrow J_1 \\ \frac{1}{n_2} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}_{\theta_0} [\psi_i(d_1) \psi_i(d_2)'] &\rightarrow J_2(d_1, d_2). \end{aligned}$$

J_1 is positive definite. When $d_1 = d_2 = d$, we write $J_2(d) := J_2(d, d)$. The pooled information matrix is defined as $J(\delta_2) = (1 - \rho)J_1 + \rho J_2(\delta_2)$.

Our asymptotic representation theorem states that any convergent sequence of

designs and policies in the finite-sample experiment converges to a design and policy in the limit experiment.

Theorem 1 (Asymptotic representation theorem): *Suppose Assumptions 1 to 5 hold. Let (δ_{2n}, c_n) be a sequence of adaptive designs and policies in the finite-sample experiment which converges in distribution under the sequence $\theta_0 + h/\sqrt{n}$ for every h . Then there exist statistics (δ_2, c) and random variables A_1, A_2, U such that*

1. $A_1 \sim N((1 - \rho)J_1h, (1 - \rho)J_1)$ independently of U , and $\delta_2 = \delta_2(A_1, U)$
2. $A_2|A_1, U \sim N(\rho J_2(\delta_2)h, \rho J_2(\delta_2))$ and $c = c(A_1, A_2, U)$
3. $(\delta_{2n}, c_n) \xrightarrow{h} (\delta_2, c)$.

Asymptotic representation theorems are often used to provide simple approximations to statistical decision problems. For the classic, non-adaptive case, see Le Cam (1972) and Van der Vaart (2000). Hirano and Porter (2009) use such a theorem to derive asymptotically optimal rules for binary treatment choice.

Hirano and Porter (2025) establish a similar result for batched multi-armed bandits. Our result, while stated for a two-wave adaptive experiment, can be extended to multiple waves. The main distinction between our result and Hirano and Porter (2025) is that we allow for continuous treatments. Our result builds on this work by extending their argument to these more complex design spaces, in which δ can determine not just the fraction of units assigned to a treatment, but also the values of continuous treatments assigned to each unit, which can depend on covariates. This generality comes at the cost of the additional Assumption 4, which is satisfied in the multi-armed bandit setting of Hirano and Porter (2025) because the score and log-likelihood are partial-sum processes. Part of our contribution is to verify this high-level condition for a class of experimental designs and models described in Appendix D and which nest our Progres application in Section 5. Finally, Hirano and Porter (2025) do not solve for optimal policies, which we consider next.

Theorem 1 applies to sequences of policies c_n that converge at the parametric rate. While this can fail in nonparametric empirical welfare maximization problems (Kitagawa and Tetenov 2018, Susan Athey and Wager 2021), it is much less restrictive in the parametric models considered here. In particular, Appendix B verifies that the Bayes-optimal policy is uniformly tight.

We note that the usefulness of Theorem 1 extends beyond the specific decision problem considered here. It does not depend on the objective function or constraints of the decision problem, nor does it impose a Bayesian framework— it is stated for fixed local alternatives. Thus, Theorem 1 can be used to analyze efficiency of estimators and tests in adaptive experiments with continuous treatments. We leave this to future work.

4.4 Asymptotic optimality

Our main result states that the procedure described in Section 3 is asymptotically optimal. We first establish a lower bound on the regret of any sequence of designs and policies in the finite-sample experiment. Then we show that the procedure described in Section 3 achieves this lower bound, and is therefore asymptotically optimal.

We first require some regularity on the decision maker’s objective and constraints.

- Assumption 6** (Nonlinear program regularity):
1. W is continuous on $\Pi \times \Theta$ and twice continuously differentiable in a neighborhood of (π_0, θ_0) . Each g_j is twice continuously differentiable in a neighborhood of π_0 .
 2. π_0 is the unique global maximizer of $W(\pi, \theta_0)$ subject to $g(\pi) \leq 0$. There exists some $\alpha < W(\pi_0, \theta_0)$, a neighborhood N of θ_0 , and a compact set C_π such that $\{\pi \in \Pi : W(\pi, \theta) \geq \alpha\} \neq \emptyset$ and $\{\pi \in \Pi : W(\pi, \theta) \geq \alpha\} \subseteq C_\pi$ for all $\theta \in N$. Π is closed and convex.
 3. The gradients $\{\nabla g_j(\pi_0) : g_j(\pi_0) = 0\}$ are linearly independent.
 4. There is a $\kappa > 0$ such that $c' \bar{H} c \geq \kappa \|c\|^2$ for all $c \in \text{span}(\mathcal{G})$ and $c' H c \geq \kappa \|c\|^2$ for all $c \in \text{span}(\Pi - \Pi)$.

These conditions are standard in the literature on sensitivity analysis for nonlinear programs under weak complementary slackness (Shapiro 1985, Bonnans and Shapiro 2013). They imply that the value of the reference decision problem (2) is locally approximated by a quadratic program. Item 3 ensures that the Lagrange multiplier λ_0 is unique, but not necessarily nonzero. This means that some constraints in the limit experiment will be inequality constraints, and the decision maker may not know whether these constraints are binding or nonbinding at the true optimal policy. Item

4 restricts our attention to strongly concave welfare functions; this rules out binary treatment choice with a linear welfare function.

We also require smoothness of the prior.

Assumption 7: *q is positive and continuous in a neighborhood of θ_0 .*

Under these assumptions, the optimal design and policy $(\delta_{2,K}^*, c_K^*)$ in the limit experiment provides a lower bound on the regret of any sequence of designs and policies in the finite-sample experiment.

Theorem 2 (Lower bound): *Suppose Assumptions 1 to 7 hold. Let $(\delta_{2,n}, c_n)$ be any sequence of designs and policies in the finite-sample experiment. Then for any fixed $K < \infty$,*

$$\liminf_{n \rightarrow \infty} \mathcal{R}_{n,K}(\delta_{2,n}, c_n) \geq \mathcal{R}_{\infty,K}^{**}.$$

Theorem 2 combines Theorem 1 with a second-order expansion of the oracle problem (2). Theorem 1 justifies the Gaussian approximation of Section 3, and is why the bound $\mathcal{R}_{\infty,K}^{**}$ is defined in a Gaussian environment. The decision problem in the limit experiment is more than just a Taylor approximation to welfare; it is a joint approximation of the objective and constraints of the decision problem. This approximation is related to existing results on sensitivity analysis of nonlinear programs under weak complementary slackness (Shapiro 1985, Bonnans and Shapiro 2013). In particular, $V(b)$ provides a second-order directional derivative of the oracle problem (2) in the direction b , up to policy-invariant centering terms. Theorem 2 combines this result with the Gaussian environment of Theorem 1 to show that the pointwise directional derivative provides an asymptotic lower bound.

We now show that our proposed experimental design $\hat{\delta}_{2,n}$ is asymptotically optimal in the sense that it attains the lower bound in Theorem 2. To establish asymptotic optimality, we construct a feasible sequence of terminal policies which attain the lower bound in Theorem 2. Let $\hat{A}_{1,n}$ and $\hat{A}_{2,n}(\delta_2)$ be the feasible analogues of $A_{1,n}$ and $A_{2,n}(\delta_2)$, where the score is evaluated at $\hat{\theta}_{1,n}$ rather than θ_0 . Let $\hat{J}_n(\delta_2) = \frac{n_1}{n} \hat{J}_{1,n} + \frac{n_2}{n} \hat{J}_{2,n}(\delta_2)$ be the feasible analogue of $J(\delta_2)$. Define the one-step estimator⁶

⁶The maximum likelihood estimator could also be used after strengthening the classical assumptions (e.g. Section 3.2 of Van der Vaart and Wellner (2013)) to ensure uniform $\sqrt{n}$ -consistency and asymptotic normality of the MLE across the set of designs. Given stochastic equicontinuity of the score process, the one-step estimator yields a simpler proof.

$\hat{\theta}_{2,n}^{\text{os}} = \hat{\theta}_{1,n} + \frac{1}{\sqrt{n}} \hat{J}_n(\hat{\delta}_{2,n})^{-1} \hat{A}_{2,n}(\hat{\delta}_{2,n})$ and $\hat{\beta}_{2,n}^{\text{os}} = \hat{\beta}_{1,n} + \hat{B}_n(\hat{\theta}_{2,n}^{\text{os}} - \hat{\theta}_{1,n})$. The localized version is $\hat{h}_n^{\text{os}} = \sqrt{n}(\hat{\theta}_{2,n}^{\text{os}} - \theta_0)$. Define

$$\hat{\pi}_n^{\text{os}} = \arg \min_{\pi \in \Pi} \hat{R}_n(\pi, \hat{\beta}_{2,n}^{\text{os}}) \quad \text{s.t.} \quad g(\pi) \leq 0$$

and let $\hat{c}_n^{\text{os}} = \sqrt{n}(\hat{\pi}_n^{\text{os}} - \pi_0)$ be the corresponding localized policy. The one-step estimator must be regular in the following sense:

Assumption 8 (Plug-in estimator): *1. The first-wave MLE is $\sqrt{n}$ consistent and interior to the parameter space.*

2. The score ψ_θ exists in a neighborhood of θ_0 and satisfies

$$\begin{aligned} \sup_{\|h\| \leq K} \left\| \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{1,n}} [\psi_{i,h}(\delta_1) - \psi_i(\delta_1)] + (1 - \rho) J_1 h \right\| &= o_{P_0}(1) \\ \sup_{d \in \Delta, \|h\| \leq K} \left\| \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} [\psi_{i,h}(d) - \psi_i(d)] + \rho J_2(d) h \right\| &= o_{P_0}(1). \end{aligned}$$

3. $\|\hat{J}_{1,n} - J_1\| = o_{P_0}(1)$ and $\sup_{d \in \Delta} \|\hat{J}_{2,n}(d) - J_2(d)\| = o_{P_0}(1)$.

4. The sequence $R_n(\hat{c}_n^{\text{os}}, h)$ is uniformly integrable under the prior $q_{\infty,K}$ for any $K < \infty$.

This assumption ensures that the feasible one-step estimator $\hat{\beta}_{2,n}^{\text{os}}$ behaves like the Gaussian limit experiment estimator $\hat{b}_{2,K}$ for large K . Interiority of $\hat{\theta}_{1,n}$ implies that the first-wave score $\hat{A}_{1,n} = 0$. Together, items 1-3 imply that the one-step process $\hat{A}_{2,n}(\cdot)$ is asymptotically equivalent to the Gaussian limit experiment score process $A_2(\cdot)$, just as Assumption 4 implies that the true second-wave score process $A_{2,n}(\cdot)$ is asymptotically equivalent to $A_2(\cdot)$. The uniform score and information conditions are implied by uniform versions of standard differentiability and dominance conditions on the likelihood.

We now show that a decision maker who uses $\hat{\delta}_{2,n}$ to design the second wave of the experiment and subsequently chooses the one-step estimator $\hat{\pi}_n^{\text{os}}$ for the final policy attains the asymptotic lower bound in Theorem 2.

Theorem 3 (Asymptotic optimality): *Suppose that for any K , Assumptions 1 to 8 hold. Then*

$$\lim_{K \rightarrow \infty} \limsup_{n \rightarrow \infty} \left\{ \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{os}) - \mathcal{R}_{\infty,K}^{**} \right\} = 0.$$

This is the main result of the paper, which justifies the use of the method proposed in Section 3. While Theorem 2 states that no feasible design and policy in the finite-sample experiment can achieve lower regret than the optimal design and policy in the limit experiment, this bound is useful only insofar as the design and policy used in practice can attain it. Theorem 3 shows that the proposed method does attain this bound. Since the limit experiment is defined under a truncated prior, the flat-prior procedure in Section 3 is not asymptotically optimal for any fixed K , since it disregards the information of the truncated prior. However, as K grows large and the truncation becomes less informative, the optimality gap between the flat-prior and truncated-prior procedure vanishes.

Previous asymptotic optimality results in adaptive experiments apply to decisions such as point estimation, hypothesis testing, or discrete treatment choice (Adusumilli 2025a,b; Hirano and Porter 2025). To our knowledge, Theorem 3 is the first asymptotic optimality result for adaptive experiments with continuous treatments, nonlinear objectives, and nonlinear constraints. A limitation of this result is that it is restricted to Bayes regret. Theorem 3 may be useful in studying asymptotic optimality under minimax regret, but we leave this to future work.

Since $\mathcal{R}_{n,K}^*(\hat{\delta}_{2,n}) \leq \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{os})$, Theorem 3 implies that a decision maker who uses $\hat{\delta}_{2,n}$ to run the experiment and then chooses the fully ex-post Bayes optimal policy $c_{n,K}^*$ will also attain the asymptotic lower bound; however, the one-step estimator is simpler to compute and is asymptotically equivalent.

5 APPLICATION TO PROGRESA

We apply our method to the Progresa cash transfer experiment in Mexico. We consider the policy choice problem of choosing the size and targeting of the cash transfer to maximize the secondary school completion rate among children in the program. When our proposed method is used to design a second wave of the Progresa experiment, we find that the expected regret of the policy choice problem is reduced by up to 70% relative to a second wave which uses the original Progresa experimental

design.

Progresa was a conditional cash transfer program administered by the Mexican government beginning in 1998. The program gave cash to households for every child enrolled in school. The size of the transfer depended on children’s gender and grade of enrollment, and the magnitude was economically significant— the ninth grade subsidy constituted about 40% of an adult male’s wage and about 66% of a child’s wage (Schultz 2004). The government conducted an experimental evaluation of the program before a larger rollout in 2000. In the original experiment, 62% of the experimental sample was treated. After observing the results of the experiment, the government enacted a policy that gave the exact same subsidy that was used in the experiment to a wider set of villages.

5.1 Model

We use a dynamic school choice model to learn about the effects of counterfactual policies that the decision maker may choose. This approach has been taken by Petra E Todd and Kenneth I Wolpin (2006) and Attanasio, Meghir, and Santiago (2012). One reason this model is useful for policy choice is that the school completion rate is a long-term outcome which is not immediately observed in the experiment. Instead, only current schooling enrollment decisions are observed. The dynamic model allows the decision maker to use the observed effects of the program to learn about the long-term effects of different policies (see Susan Athey, Chetty, et al. (2025) for a related but less structural approach to this problem).

In describing the model, we suppress the unit index i . At each age τ before age 18, households may choose whether a child in grade s_τ will attend school, denoted $y_\tau = 1$ or $y_\tau = 0$. If the child attends school, they receive a subsidy z_{s_τ} which depends on the grade of enrollment. The utility a household receives from a child enrolling in school depends on the subsidy z as well as the child’s age τ , current grade level s_τ , a female sex indicator f , and an indicator for attending secondary school. The difference at age τ in the utility from enrolling in school versus not enrolling is given by

$$u_\tau(s, f, z) = \gamma_0 + \gamma_1\tau + \gamma_2s + \gamma_3f + \gamma_4\mathbb{1}[s \geq 7] + \gamma_5(1 - f)z + \gamma_6(1 - f)\tau z + \gamma_7fz + \gamma_8f\tau z + \epsilon_\tau.$$

Households choose a sequence of school attendance y_τ to maximize long-term utility. Each period, if the child was enrolled in school they may advance a grade with probability $p_{\text{pass}}(\tau, s_\tau)$ depending on age and grade, or they may fail and remain at the same grade. Failure is assumed to be exogenous. Finally, at age $\tau = 18$ households obtain a terminal value

$$v_{18} = \gamma_9 \mathbb{1}[s_{18} \geq 10]$$

which depends on whether the child completed secondary school. Thus, households solve

$$\begin{aligned} \max_{\{y_\tau\}_{\tau=6}^{17}} \sum_{\tau=6}^{17} \eta^\tau \mathbb{E}[y_\tau u_\tau(s_\tau, f, z_{s_\tau})] + \eta^{18} \mathbb{E}[v_{18}] \\ s_{\tau+1} = s_\tau + y_\tau \text{Bernoulli}(p_{\text{pass}}(\tau, s_\tau)) \end{aligned}$$

where η is a calibrated discount factor and households may observe ϵ_τ before choosing y_τ . We assume that ϵ_τ has a logistic distribution, which enables us to compute the value function for the households by backwards induction. We can estimate $\theta = (\gamma_0, \dots, \gamma_9)$ by maximum likelihood from a cross-section of children so long as there is sufficient variation in the subsidy for the model to be identified.

5.2 Decision problem

The decision problem is to choose (i) an experimental design and (ii) a policy that maximizes the fraction of children graduating from secondary school. A policy π is a four-dimensional vector specifying the subsidy amounts z_s for boys and girls in primary and secondary school. The welfare function is

$$W(\pi, \theta) = P_{\pi, \theta}[s_{18} \geq 9]$$

where $P_{\pi, \theta}$ represents the probability distribution of s_{18} when θ is the true parameter value and subsidy amounts are determined by π . Other welfare functions, including those reflecting inequality aversion between genders, are also compatible with our method.

We suppose the decision maker has access to a first wave of experimental data and wants to run a second wave to help choose the optimal policy. The first wave consists

of a sample of n_1 children from the original Progresas experiment. The design of the second wave can differ from the original Progresas experiment in two ways. First, the decision maker can change the peso amount of the subsidy offered to boys and girls in each grade. Second, the decision maker can let the probability of treatment vary by gender. Thus, the design δ_2 is a six-dimensional vector describing the amount of the subsidy offered to each group and the probability of being treated for boys and girls. We consider $n_1 = n_2 \in \{250, 500, 1000\}$.

Both the experimental subsidy and the chosen policy subsidy are subject to constraints. First, the total amount of the chosen subsidy cannot exceed the total amount of the original Progresas subsidy, although the decision maker can allocate the subsidy differently across gender and the grade of enrollment. The decision maker may also allocate the subsidy differently across the treatment and control groups in the experiment, but the treated-share-weighted average of the experimental subsidy cannot exceed that of the Progresas experiment. Second, the subsidy must be nonnegative, preventing the decision maker from taxing children in some grades to subsidize children in other grades. We find that both of these constraints bind in the optimal experiment and policy, although the decision maker may be unsure about which grades will have strictly positive subsidies. We also constrain the subsidy in each grade to be less than 1000 pesos.

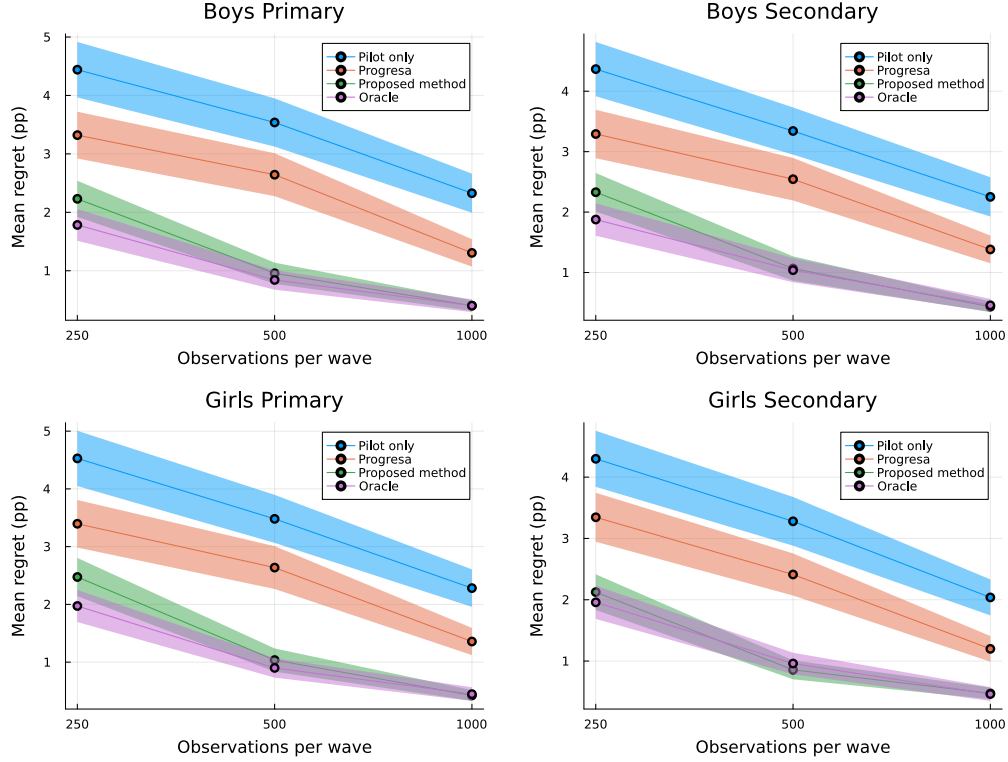
5.3 Optimal experiment

We now present the result of applying our proposed experimental design to this decision problem. Implementation details are provided in Appendix E.

Following Section 4, we focus on the performance of the proposed method in statistically challenging settings where the optimal policy and the active set of constraints are difficult to learn. Specifically, we evaluate the expected regret of our method when the true parameter is in a neighborhood of a point θ_0 where (i) the optimal policy π_0 is a vertex of the feasible set and (ii) the nonnegativity constraints are weakly binding, so the decision maker is unsure which groups should receive a positive subsidy. We calibrate these points by estimating θ_0 by maximum likelihood on a large holdout set, subject to the constraints that the KKT conditions of the policy choice problem hold with $\lambda_{j0} = 0$ for the nonnegativity constraints j . This yields four points θ_0 corresponding to the four vertices of the feasible set that lie on the budget constraint.

For each sample size and each θ_0 estimated on the holdout set, we run a Monte

Figure 1: Expected regret by experimental design



Note: Each panel shows expected regret and 95% Monte Carlo confidence intervals when θ is local to one of four reference points θ_0 corresponding to four vertices of the feasible set. Each vertex is characterized by a different group of children receiving a positive subsidy. Not all replications achieved convergence; expected regret is computed based on the $> 99\%$ of replications that converged for each vertex and sample size.

Carlo simulation to compute the expected regret of the proposed method. We sample θ from a Uniform distribution on a neighborhood of θ_0 and generate a first wave of experimental data from the original Progres experiment which is used to estimate the optimal design following Section 3. We then generate a second wave of experimental data using the proposed design. For comparison, we also generate a second wave of experimental data using the original Progres design and a second wave using an oracle optimal design which uses the true θ and Fisher information to choose the optimal design, rather than using first-wave plugins for these quantities. We then estimate the optimal policy using both waves from each design and compute the resulting welfare.

Figure 1 shows the headline expected regret results. More detailed results are

provided in Appendix E. We compare the expected regret of the proposed method to the expected regret of optimal policies estimated using data generated by alternative experimental designs. “Pilot only” refers to the expected regret of $\hat{\pi}_{1,n}$, which disregards the second wave. “Progresa” refers to the expected regret of $\hat{\pi}_{2,n}$ when the second wave is generated using the original Progresa design, just like the first wave. “Estimated optimal” refers to the expected regret of $\hat{\pi}_{2,n}$ when the second wave is generated using the proposed design $\hat{\delta}_{2,n}$. “Oracle optimal” refers to the expected regret of $\hat{\pi}_{2,n}$ when the second wave is generated using an oracle version of the proposed design in which the Gaussian and quadratic approximations are known exactly rather than estimated.

Across all sample sizes and reference points, the proposed method substantially reduces expected regret relative to the original Progresa design. The gain is particularly large for $n_1 = n_2 \in \{500, 1000\}$, in which regret is reduced by 50-70%. The gains at $n_1 = n_2 = 250$ are smaller, suggesting that sampling error in the first wave is enough to degrade the quality of the estimated design, but even this noisily estimated design is more valuable than doubling the sample size of the original Progresa design. We use the same method to estimate the optimal policy in each experimental design, so this difference is due to the value of the dataset generated by different designs.

Appendix E provides additional details on how the proposed design differs from the original Progresa design and how these differences affect the quality of the chosen policy. The optimal design tends to focus exclusively on secondary school and give large experimental subsidies to a small treatment group. This is despite θ_0 being chosen so that the marginal effect of increasing the subsidy is equalized across all four groups, so that allocating the subsidy to primary school children is optimal for some values of θ . This is therefore not an artifact of primary school being unimportant for the policy choice (in our model, a large coefficient on years of education can make early-year subsidies important for eventual graduation), but rather a reflection of how informative different experimental subsidies are for learning about the optimal policy, which is jointly determined by the objective, constraints, and information of the model. In turn, the more informative designs improve the quality of the chosen policy by increasing the probability of allocating more subsidy to the group with the highest response to the subsidy.

The optimal experiment is more effective for two reasons. First, the optimal experiment focuses on learning about the effects of counterfactual policies through the

marginal effect of policy changes. To choose optimal policies, the decision maker only needs to know the effect of increasing the subsidy along each dimension of the policy. Second, the optimal experiment takes into account policy constraints to determine which of these marginal effects are most important to learn about. By focusing the experiment on learning about the decision-relevant marginal effects of policy changes, the optimal experiment is able to more effectively inform policy choice.

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A PROOF OF THEOREM 1

Let $\mathcal{D} = \mathcal{Y} \times \mathcal{Z}$ be the sample space for the random data across all units. Let $\Omega_n = \mathcal{D}^{n_1} \times \mathcal{D}^{n_2} \times [0, 1]$ be the sample space of the finite-sample experiment, along with an auxiliary uniform random variable U . The statistics δ_{2n}, c_n are measurable maps on Ω_n .

Let $\phi_n : \mathcal{D}^{n_1} \times [0, 1] \rightarrow \Delta$ be a sequence of adaptive experimental design rules, so that $\delta_{2n} = \phi_n(D_{1,n}, U)$. Given a sequence of adaptive designs ϕ_n , we construct a sequence of probability measures $P_{n,h}^{\phi_n}$ on Ω_n as follows. Given our potential outcomes model and the law $P_{\epsilon,\nu}$ of the structural error term and randomization device, define the conditional law of the second wave as

$$\begin{aligned} & P_{2,n,h}^{\phi_n}(dD_{2,n} \mid D_{1,n}, U) \\ &= \prod_{i \in \mathcal{I}_{2,n}} P_{\epsilon,\nu} \left(dy \left(z(x_i, \nu_i, \phi_n(D_{1,n}, U)), x_i, \epsilon_i; \theta_0 + \frac{h}{\sqrt{n}} \right), dz(x_i, \nu_i; \phi_n(D_{1,n}, U)) \right) \end{aligned}$$

and the full law of the finite-sample experiment as

$$P_{n,h}^{\phi_n}(dU, dD_{1,n}, dD_{2,n}) = \mu(dU) \times P_{1,n,h}(dD_{1,n}) \times P_{2,n,h}^{\phi_n}(dD_{2,n} \mid D_{1,n}, U).$$

The law $P_{n,h}^{\phi_n}$ reflects that the adaptive structure induces dependence between the first and second wave of the experiment.

In this appendix, we use the superscript ϕ_n on probabilities and expectations in the finite-sample experiment when those quantities depend on the adaptive design rule. We omit it when $d \in \Delta$ is fixed or the quantity does not otherwise depend on the design rule.

Lemma 1: *Under Assumptions 1, 2, and 3, the score satisfies the Lindeberg condition*

$$\frac{1}{n} \sum_{i \in \mathcal{I}_{1,n}} \mathbb{E}_0 [\|\psi_i(\delta_1)\|^2 \mathbb{1}[\|\psi_i(\delta_1)\| > \varepsilon \sqrt{n}]] \rightarrow 0$$

$$\frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}_0 [\|\psi_i(d)\|^2 \mathbb{1}[\|\psi_i(d)\| > \varepsilon \sqrt{n}]] \rightarrow 0$$

for every $d \in \Delta$ and $\varepsilon > 0$.

Proof. We prove the result for the second wave; the first is identical with a fixed design. Fix a direction $v \in \mathbb{R}^\ell$ and define $S = v' \psi(y \mid z, x)$. Define the DQM remainder

$$\omega_{z,x}(h) = \sqrt{p_{y|z,x}(y \mid z, x; \theta_0 + h)} - \sqrt{p_{y|z,x}(y \mid z, x; \theta_0)} - \frac{1}{2} h' \psi(y \mid z, x) \sqrt{p_{y|z,x}(y \mid z, x; \theta_0)}.$$

Fix $t > 0$. On the event $S > 4/t$, $(t/2)S - 1 \geq (t/4)S$. Nonnegativity of densities implies that the DQM remainder satisfies

$$\omega_{z,x}(-tv) = \sqrt{p_{y|z,x}(y \mid z, x; \theta_0 - tv)} + \left(-1 + \frac{t}{2}S\right) \sqrt{p_{y|z,x}(y \mid z, x; \theta_0)} \geq \frac{t}{4}|S| \sqrt{p_{y|z,x}(y \mid z, x; \theta_0)}.$$

Thus,

$$\|\omega_{z,x}(-tv)\|^2 \geq \frac{t^2}{16} \mathbb{E}_0 [\|S\|^2 \mathbb{1}[S > 4/t]].$$

Likewise, by considering the event $S < -4/t$, we have

$$\|\omega_{z,x}(tv)\|^2 \geq \frac{t^2}{16} \mathbb{E}_0 [\|S\|^2 \mathbb{1}[S < -4/t]]$$

and therefore

$$\sup_{z,x} \mathbb{E}_0 [\|S\|^2 \mathbb{1}[\|S\| > 4/t]] \leq \frac{16}{t^2} \left(\sup_{z,x} \|\omega_{z,x}(-tv)\|^2 + \sup_{z,x} \|\omega_{z,x}(tv)\|^2 \right)$$

which is $o(1)$ by Assumption 2. Since the score is finite-dimensional, this directional bound implies that for any $d \in \Delta$, $\mathbb{E}_0 [\|\psi_i(d)\|^2 \mathbb{1}[\|\psi_i(d)\| > \varepsilon \sqrt{n}]] \rightarrow 0$ uniformly in z, x and the result follows. $\square$

Lemma 2: Suppose Assumptions 1, 2, 3, and 5 hold. Then for any fixed $d \in \Delta$ and any h ,

$$\sum_{i \in \mathcal{I}_{2,n}} \log \frac{p_{y|z,x}(y_i(d) \mid z_i(d), x_i; \theta_0 + h/\sqrt{n})}{p_{y|z,x}(y_i(d) \mid z_i(d), x_i; \theta_0)} = \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} h' \psi_i(d) - \frac{1}{2} h' \rho J_2(d) h + o_{P_0}(1)$$

Proof. In this proof, we will use the shorthand $p_n(\cdot \mid \cdot, \cdot) := p_{y|z,x}(\cdot \mid \cdot, \cdot; \theta_0 + h/\sqrt{n})$ and $p(\cdot \mid \cdot, \cdot) := p_{y|z,x}(\cdot \mid \cdot, \cdot; \theta_0)$. We will also suppress the dependence of y_i and z_i on d to aid readability.

Define the random variable $S_{n,i} = 2(\sqrt{p_n/p}(y_i \mid z_i, x_i) - 1)$ which is well-defined P_0 -almost everywhere. We write a Taylor expansion of the log-likelihood ratio in terms of S_{ni} :

$$\sum_{i \in \mathcal{I}_{2,n}} \log \frac{p_n}{p}(y_i \mid z_i, x_i) = 2 \sum_{i \in \mathcal{I}_{2,n}} \log \left(1 + \frac{1}{2} S_{ni} \right) = \sum_{i \in \mathcal{I}_{2,n}} \left(S_{ni} - \frac{1}{4} S_{ni}^2 + \frac{1}{2} S_{ni}^2 \text{rem}(S_{ni}) \right) \quad (4)$$

where $\text{rem}(s) \rightarrow 0$ as $s \rightarrow 0$.

First, observe that

$$\begin{aligned} \mathbb{E} \left[\sum_{i \in \mathcal{I}_{2,n}} S_{ni} \right] &= - \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E} \left[\int (\sqrt{p_n} - \sqrt{p})^2 d\mu \right] \\ &= - \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E} \left[\int \left(\frac{1}{2\sqrt{n}} h' \psi_i(d) \sqrt{p} \right)^2 d\mu + s_n(z_i, x_i) \right] \\ &\rightarrow -\frac{1}{4} h' \rho J_2(d) h \end{aligned}$$

where the last line follows from Assumptions [1](#), [2](#), and [5](#). Next, observe that

$$\begin{aligned} \text{Var} \left(\sum_{i \in \mathcal{I}_{2,n}} S_{ni} - \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} h' \psi_i(d) \right) &\leq \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E} \left[\left(S_{ni} - \frac{1}{\sqrt{n}} h' \psi_i(d) \right)^2 \right] \\ &= 4 \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E} \left[\int \left(\sqrt{p_n} - \sqrt{p} - \frac{1}{2\sqrt{n}} h' \psi_i(d) \sqrt{p} \right)^2 d\mu \right] \\ &\rightarrow 0 \end{aligned}$$

again by DQM. Combining the previous two displays, we have that the mean and variance of

$$\sum_{i \in \mathcal{I}_{2,n}} S_{ni} - \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} h' \psi_i(d) + \frac{1}{4} h' \rho J_2(d) h \rightarrow 0$$

and hence this expression converges to zero in probability.

Now define $U_{ni} = \sqrt{n}S_{ni} - h'\psi_i(d)$. We have that

$$\begin{aligned} & \mathbb{E} \left| \frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} (2h'\psi_i(d)U_{ni} + U_{ni}^2) \right| \\ & \leq 2 \left(\frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}(h'\psi_i(d))^2 \right)^{1/2} \left(\frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}U_{ni}^2 \right)^{1/2} + \frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}U_{ni}^2 \\ & = O(1)o(1) + o(1) \end{aligned}$$

by the preceding argument. Since $nS_{ni}^2 = (h'\psi_i(d))^2 + 2h'\psi_i(d)U_{ni} + U_{ni}^2$,

$$\begin{aligned} \sum_{i \in \mathcal{I}_{2,n}} S_{ni}^2 &= \frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} (h'\psi_i(d))^2 + o_{P_0}(1) \\ &= \rho h' J_2(d) h + o_{P_0}(1) \end{aligned}$$

by Assumption 1, Assumption 5, and the Lindeberg law of large numbers, which applies to the scores by Lemma 1.

Combining the preceding calculations, we have that

$$\sum_{i \in \mathcal{I}_{2,n}} \left(S_{ni} - \frac{1}{4} S_{ni}^2 \right) = \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} h'\psi_i(d) - \frac{1}{2} h' \rho J_2(d) h + o_{P_0}(1).$$

To complete the proof, it remains to show that the remainder term in (4) is $o_{P_0}(1)$. Observe that

$$\sum_{i \in \mathcal{I}_{2,n}} S_{ni}^2 \text{rem}(S_{ni}) \leq \max_{i \in \mathcal{I}_{2,n}} |\text{rem}(S_{ni})| \sum_{i \in \mathcal{I}_{2,n}} S_{ni}^2$$

where $\sum_{i \in \mathcal{I}_{2,n}} S_{ni}^2$ is $O_{P_0}(1)$, so we want to show that $\max_{i \in \mathcal{I}_{2,n}} |\text{rem}(S_{ni})| = o_{P_0}(1)$. For any $\varepsilon > 0$ we have

$$P \left(\max_{i \in \mathcal{I}_{2,n}} |S_{ni}| > \varepsilon \right) \leq \sum_{i \in \mathcal{I}_{2,n}} P(|S_{ni}| > \varepsilon).$$

Each term in the sum obeys

$$\begin{aligned}
P(|S_{ni}| > \varepsilon) &\leq P\left((h'\psi_i(d))^2 > \frac{1}{4}n\varepsilon^2\right) + P\left(U_{ni}^2 > \frac{1}{4}n\varepsilon^2\right) \\
&= P\left((h'\psi_i(d))^2 \mathbb{1}[(h'\psi_i(d))^2 > \frac{1}{4}n\varepsilon^2] > \frac{1}{4}n\varepsilon^2\right) + P(U_{ni}^2 > \frac{1}{4}n\varepsilon^2) \\
&\leq 4n^{-1}\varepsilon^{-2}\mathbb{E}\left[(h'\psi_i(d))^2 \mathbb{1}[(h'\psi_i(d))^2 > \frac{1}{4}n\varepsilon^2]\right] + 4n^{-1}\varepsilon^{-2}\mathbb{E}[U_{ni}^2].
\end{aligned}$$

By Lemma 1,

$$\sum_{i \in \mathcal{I}_{2,n}} 4n^{-1}\varepsilon^{-2}\mathbb{E}\left[(h'\psi_i(d))^2 \mathbb{1}[(h'\psi_i(d))^2 > \frac{1}{4}n\varepsilon^2]\right] \rightarrow 0$$

and moreover

$$\sum_{i \in \mathcal{I}_{2,n}} 4n^{-1}\varepsilon^{-2}\mathbb{E}[U_{ni}^2] \rightarrow 0.$$

This shows that $\max_{i \in \mathcal{I}_{2,n}} |\text{rem}(S_{ni})| = o_{P_0}(1)$ and the last term in the Taylor expansion is $o_{P_0}(1)O_{P_0}(1) = o_{P_0}(1)$. $\square$

Lemma 3: *Suppose Assumptions 1 to 5 hold. Then every subsequence has a further subsequence along which*

$$(A_{1,n}, A_{2,n}(\cdot)) \xrightarrow{0} (A_{1,\infty}^0, A_{2,\infty}^0(\cdot))$$

where $A_{1,\infty}^0 \sim N(0, (1-\rho)J_1)$ and $A_{2,\infty}^0$ is a Gaussian process with mean zero and covariance kernel $\rho J_2(\cdot, \cdot)$, independent of $A_{1,\infty}^0$ and $\delta_{2,\infty}^0$. Along this subsequence,

$$\log \frac{dP_{n,h}^{\phi_n}}{dP_{n,0}^{\phi_n}} \xrightarrow{0} h'A_{1,\infty}^0 - \frac{1}{2}h'(1-\rho)J_1h + h'A_{2,\infty}^0(\delta_{2,\infty}^0) - \frac{1}{2}h'\rho J_2(\delta_{2,\infty}^0)h =: \Lambda_h$$

for any h . Moreover, $P_{n,h}^{\phi_n}$ is contiguous to $P_{n,0}^{\phi_n}$ along the full sequence.

Proof. Assumptions 1, 5, Lemma 1 and the Lindeberg-Feller CLT implies finite-dimensional convergence of the score process. Assumption 4 then implies that

$$(A_{1,n}, A_{2,n}(\cdot)) \xrightarrow{0} (A_{1,\infty}^0, A_{2,\infty}^0(\cdot))$$

where $A_{1,\infty}^0 \sim N(0, (1 - \rho)J_1)$ and $A_{2,\infty}^0(\cdot)$ is a Gaussian process with mean zero and covariance kernel $\rho J_2(\cdot, \cdot)$, independent of $A_{1,\infty}^0$. In particular, the covariance kernel of $A_{2,\infty}^0(\cdot)$ must be continuous. Since Δ is compact, the random elements $(A_{1,n}, A_{2,n}(\cdot), \delta_{2,n})$ are jointly uniformly tight and therefore converge in distribution along a subsequence to some limit $(A_{1,\infty}^0, A_{2,\infty}^0(\cdot), \delta_{2,\infty}^0)$.

The likelihood ratio may be written as

$$\frac{P_{n,h}^{\phi_n}}{P_{n,0}^{\phi_n}}(dU, dD_{1,n}, dD_{2,n}) = \frac{P_{1,n,h}}{P_{1,n,0}}(dD_{1,n}) \times \frac{P_{2,n,h}^{\phi_n}}{P_{2,n,0}^{\phi_n}}(dD_{2,n} \mid D_{1,n}, U),$$

where h -independent terms cancel in the likelihood ratio. In particular,

$$\frac{P_{2,n,h}^{\phi_n}}{P_{2,n,0}^{\phi_n}}(dD_{2,n} \mid D_{1,n}, U) = \prod_{i \in \mathcal{I}_{2,n}} \frac{p_{y|z,x}(y_i(\phi_n(D_{1,n}, U)) \mid z_i(\phi_n(D_{1,n}, U)), x_i; \theta_0 + h/\sqrt{n})}{p_{y|z,x}(y_i(\phi_n(D_{1,n}, U)) \mid z_i(\phi_n(D_{1,n}, U)), x_i; \theta_0)}.$$

By Lemma 2, we have that

$$\sum_{i \in \mathcal{I}_{2,n}} \log \frac{p_{y|z,x}(y_i(d) \mid z_i(d), x_i; \theta_0 + h/\sqrt{n})}{p_{y|z,x}(y_i(d) \mid z_i(d), x_i; \theta_0)} = h' A_{2,n}(d) - \frac{1}{2} h' \rho J_2(d) h + o_{P_0}(1)$$

pointwise in d . By Assumption 4 and continuity of $J_2(\cdot)$, both the left and right-hand sides are stochastically equicontinuous in d and therefore the convergence is uniform in d . A similar expansion holds for the first wave of the experiment, where δ_1 is fixed.

Since $J_2(\cdot, \cdot)$ is continuous and $A_{2,\infty}^0(\cdot)$ has almost surely continuous sample paths, we can apply the extended continuous mapping theorem to conclude that

$$\log \frac{dP_{n,h}^{\phi_n}}{dP_{n,0}^{\phi_n}} \xrightarrow{0} h' A_{1,\infty}^0 - \frac{1}{2} h'(1 - \rho) J_1 h + h' A_{2,\infty}^0(\delta_{2,\infty}^0) - \frac{1}{2} h' \rho J_2(\delta_{2,\infty}^0) h$$

along this subsequence.

To verify contiguity along this subsequence, it suffices to show that $\mathbb{E}_0[\exp(\Lambda_h)] = 1$ by Le Cam's first lemma (Theorem 6.4 of Van der Vaart (2000)). By the law of iterated expectations,

$$\mathbb{E}_0[\exp(\Lambda_h)] = \mathbb{E}_0 \left[\exp \left(h' A_{1,\infty}^0 - \frac{1}{2} h'(1 - \rho) J_1 h \right) \right]$$

$$\times \mathbb{E}_0 \left[\exp \left(h' A_{2,\infty}^0(\delta_{2,\infty}^0) - \frac{1}{2} h' \rho J_2(\delta_{2,\infty}^0) h \right) \mid A_{1,\infty}^0, \delta_{2,\infty}^0 \right].$$

Since the process $A_{2,\infty}^0(\cdot)$ is independent of $A_{1,\infty}^0$ and $\delta_{2,\infty}^0$, $A_{2,\infty}^0(\delta_{2,\infty}^0) \sim N(0, \rho J_2(\delta_{2,\infty}^0))$ conditional on $\delta_{2,\infty}^0$. Thus, the inner conditional expectation is 1 almost surely by properties of log-normal random variables, and likewise the outer expectation. This establishes contiguity along the subsequence, and since the subsequence was arbitrary, contiguity holds along the full sequence. $\square$

We can now prove Theorem 1.

Proof of Theorem 1. By Lemma 3, $P_{n,h}^{\phi_n}$ is contiguous to $P_{n,0}^{\phi_n}$ and $\log \left(dP_{n,h}^{\phi_n} / dP_{n,0}^{\phi_n} \right) \xrightarrow{0} \Lambda_h$ where the law of Λ_h is given by

$$P_0(\Lambda_h \in E) = P_0 \left(h' A_{1,\infty}^0 - \frac{1}{2} h'(1 - \rho) J_1 h + h' A_{2,\infty}^0(\delta_{2,\infty}^0) - \frac{1}{2} h' \rho J_2(\delta_{2,\infty}^0) h \in E \right)$$

where $(A_{1,\infty}^0, A_{2,\infty}^0(\cdot), \delta_{2,\infty}^0)$ are as in Lemma 3. Since (δ_{2n}, c_n) is tight under $h = 0$ by assumption, and $A_{1,n}$ and $A_{2,n}(\cdot)$ are tight by Lemma 3, $(\delta_{2n}, c_n, A_{1,n}, A_{2,n}(\cdot), \log(dP_{n,h}^{\phi_n} / dP_{n,0}^{\phi_n}))$ is jointly tight under $h = 0$. Thus, by Prohorov's theorem (Van der Vaart (2000) Theorem 2.4), there exists a subsequence along which

$$\left(\delta_{2n}, c_n, A_{1,n}, A_{2,n}(\cdot), \log \frac{dP_{n,h}^{\phi_n}}{dP_{n,0}^{\phi_n}} \right) \xrightarrow{0} (\delta_\infty^0, c_\infty^0, A_{1,\infty}^0, A_{2,\infty}^0, \Lambda_h) \quad (5)$$

for every h . The log-likelihood ratio in (5) is between two fixed sequences of probability measures that are contiguous. Moreover, the law of Λ_h is fixed—it is not a function of the realized random design $\delta_{2,\infty}^0$. The law of Λ_h does depend on the fixed sequence of adaptive design rules ϕ_n , which are maps from $\mathcal{D}^{n_1} \times [0, 1]$ to Δ . However, the joint convergence in (5) in conjunction with Lemma 3 implies that

$$\Lambda_h = h' A_{1,\infty}^0 - \frac{1}{2} h'(1 - \rho) J_1 h + h' A_{2,\infty}^0(\delta_{2,\infty}^0) - \frac{1}{2} h' \rho J_2(\delta_{2,\infty}^0) h$$

almost surely, giving a representation of the log-likelihood ratio in terms of the limiting random elements.

Thus, we can apply Le Cam's third lemma (Van der Vaart (2000) Theorem 6.6) to conclude that the function $L_h(E) = \mathbb{E}_0 [\mathbb{1}[(\delta_\infty^0, c_\infty^0) \in E] \exp(\Lambda_h)]$ defines a probability measure and $(\delta_{2n}, c_n) \xrightarrow{h} L_h$ along the subsequence for which the convergence in (5)

holds. Moreover, since the marginal sequence $(\delta_{2,n}, c_n)$ converges in distribution under h along the full sequence, L_h must be the limiting distribution of $(\delta_{2,n}, c_n)$ under h as well.

Having derived the limiting distribution of $(\delta_{2,n}, c_n)$ under every h , we now construct a statistic in a Gaussian environment which has this distribution for every h . The construction is similar to the proof of Theorem 3 in Hirano and Porter (2025) and uses conditional vector quantile functions (Carlier, Chernozhukov, and Galichon 2016).

Let A_1 be a Gaussian random vector with mean $(1 - \rho)J_1 h$ and covariance matrix $(1 - \rho)J_1$, $A_2(\cdot)$ be a Gaussian process with mean $\rho J_2(\cdot)h$ and covariance function $\rho J_2(\cdot, \cdot)$, and U be a uniform random variable on $[0, 1]$ independent of $(A_1, A_2(\cdot))$. Let $(\delta_{2,\infty}^h, c_\infty^h)$ be the weak limit of $(\delta_{2,n}, c_n)$ under h , which exist by assumption. Our goal is to construct statistics (δ_2, c) where $\delta_2 = \phi(A_1, U)$ and $c = C(A_1, A_2(\phi(A_1, U)), U)$ such that for any h , $(\delta_2, c) \stackrel{h}{\sim} (\delta_{2,\infty}^h, c_\infty^h)$. Let $Q_\delta(u \mid a_1)$ be the conditional vector quantile function of $\delta_{2,\infty}^0$ given $A_{1,\infty}^0 = a_1$. Let $Q_c(u \mid a_1, d, a_2)$ be the conditional vector quantile function of c_∞^0 given $A_{1,\infty}^0 = a_1$, $\delta_{2,\infty}^0 = d$, and $A_{2,\infty}^0(d) = a_2$. Note that $(A_{1,\infty}^0, A_{2,\infty}^0(\cdot), U)$ has the same distribution as $(A_1, A_2(\cdot), U)$ when $h = 0$.

From U , construct uniform random vectors U_1 and U_2 on the unit cube that are independent of each other and of $(A_1, A_2(\cdot))$. Define

$$\delta_2 = Q_\delta(U_1 \mid A_1) \quad c = Q_c(U_2 \mid A_1, \delta_2, A_2(\delta_2)).$$

Then for $h = 0$, it follows from the definition of conditional vector quantile functions that $(\delta_2, c) \stackrel{0}{\sim} (\delta_{2,\infty}^0, c_\infty^0)$ and therefore $(\delta_{2,n}, c_n) \stackrel{0}{\rightsquigarrow} (\delta_2, c)$. Thus, $\delta_2 = \phi(A_1, U_1) = Q_\delta(U_1 \mid A_1)$ is the limiting design rule for the second wave of the experiment.

We now verify that the statistic (δ_2, c) has the desired distribution under $h \neq 0$ as well. For some values of d in Δ , $J_2(d)$ may be singular, and therefore $A_2(d)$ may not admit a density with respect to Lebesgue measure on $\mathbb{R}^\ell$. However, it admits a density with respect to Lebesgue measure on the support of $A_2(d)$, which is $\mathcal{A}(d) = \text{range}(J_2(d))$. This density is given by

$$\frac{1}{\sqrt{\det^\dagger(2\pi\rho J_2(d))}} \exp\left(-\frac{1}{2}(a - \rho J_2(d)h)'(\rho J_2(d))^\dagger(a - \rho J_2(d)h)\right)$$

where $\det^\dagger$ denotes the pseudo-determinant and $(\rho J_2(d))^\dagger$ denotes the Moore-Penrose

pseudo-inverse of $\rho J_2(d)$.

Let E_1, E_2 be Borel sets in Δ and $\mathbb{R}^k$ respectively. Then

$$\begin{aligned} & P_h[\delta_2 \in E_1, c \in E_2] \\ &= \mathbb{E}_h [\mathbb{1} [Q_\delta(U_1 \mid A_1) \in E_1, Q_c(U_2 \mid A_1, Q_\delta(U_1 \mid A_1)), A_2(Q_\delta(U_1 \mid A_1))) \in E_2]] \\ &= \mathbb{E}_0 [\mathbb{E}_h [\mathbb{1} [Q_\delta(U_1 \mid A_1) \in E_1, Q_c(U_2 \mid A_1, Q_\delta(U_1 \mid A_1)), A_2(Q_\delta(U_1 \mid A_1))) \in E_2] \mid U_1, U_2]] \end{aligned}$$

where the second equality uses the law of iterated expectations and the fact that the distribution of U_1 and U_2 does not depend on h . Using the formula for the densities of A_1 and $A_2(d)$, this is equal to

$$\begin{aligned} & \mathbb{E}_0 \left[\int_{\mathbb{R}^\ell} \int_{\mathcal{A}(Q_\delta(U_1 \mid a_1))} \mathbb{1} [Q_\delta(U_1 \mid a_1) \in E_1, Q_c(U_2 \mid a_1, Q_\delta(U_1 \mid a_1)), a_2] \in E_2] \right. \\ & \quad \times \exp \left(h' a_1 - \frac{1}{2} h' (1 - \rho) J_1 h \right) \exp \left(h' a_2 - \frac{1}{2} h' \rho J_2(Q_\delta(U_1 \mid a_1)) h \right) \\ & \quad \times \frac{1}{\sqrt{\det(2\pi(1 - \rho)J_1)}} \exp \left(-\frac{1}{2} a_1' ((1 - \rho)J_1)^{-1} a_1 \right) \\ & \quad \times \frac{1}{\sqrt{\det^\dagger(2\pi\rho J_2(Q_\delta(U_1 \mid a_1)))}} \exp \left(-\frac{1}{2} a_2' (\rho J_2(Q_\delta(U_1 \mid a_1)))^\dagger a_2 \right) da_2 da_1 \Big] \\ &= \mathbb{E}_0 \left[\mathbb{1} [Q_\delta(U_1 \mid A_1) \in E_1, Q_c(U_2 \mid A_1, Q_\delta(U_1 \mid A_1)), A_2(Q_\delta(U_1 \mid A_1))) \in E_2] \right. \\ & \quad \times \exp \left(h' A_1 - \frac{1}{2} h' (1 - \rho) J_1 h \right) \exp \left(h' A_2(Q_\delta(U_1 \mid A_1)) - \frac{1}{2} h' \rho J_2(Q_\delta(U_1 \mid A_1)) h \right) \Big] \\ &= \mathbb{E}_0 \left[\mathbb{1} [(\delta_2, c) \in E_1 \times E_2] \right. \\ & \quad \times \exp \left(h' A_1 - \frac{1}{2} h' (1 - \rho) J_1 h \right) \exp \left(h' A_2(\delta_2) - \frac{1}{2} h' \rho J_2(Q_\delta(U_1 \mid A_1)) h \right) \Big] \end{aligned}$$

where the first expression uses the formula for the densities of A_1 and $A_2(d)$ under h , the second uses the fact that $J_2(d)J_2(d)^\dagger a = a$ for any $d \in \Delta$ and $a \in \mathcal{A}(d)$, the third uses the formula for the densities of A_1 and $A_2(d)$ under $h = 0$, and the fourth uses the definitions of δ_2 and c . The resulting expression is $L_h(E_1 \times E_2)$, and therefore for any h we have $(\delta_{2,n}, c_n) \xrightarrow{h} (\delta_2, c)$. $\square$

We now state a corollary that allows us to apply the conclusions of Theorem 1 without directly assuming the convergence holds under any h .

Corollary 1: *Suppose Assumptions 1 to 5 hold. If $(\delta_{2,n}, c_n)$ is tight under $h = 0$, then every subsequence has a further subsequence along which $(\delta_{2,n}, c_n) \xrightarrow{h} (\delta_2, c)$ for any h and (δ_2, c) is a design/policy in the limit experiment.*

Proof. Extract a subsequence along which $(\delta_{2,n}, c_n, A_{1,n}, A_{2,n}(\cdot))$ converges weakly under $h = 0$. Along this subsequence, the convergence in (5) holds for any h by the argument in the proof of Theorem 1, since this argument does not use the weak convergence of $(\delta_{2,n}, c_n)$ under h . Then we can use Le Cam's third lemma to conclude that $(\delta_{2,n}, c_n) \xrightarrow{h} L_h$ along this subsequence for any h . Hence, the conditions of Theorem 1 hold along this subsequence and the conclusion follows. $\square$

B PROOF OF THEOREM 2

For the remainder of the proofs, we suppress the superscript ϕ_n on $\mathbb{E}_h$ and P_h indicating the dependence of the data on the adaptive design rule. In the finite sample experiment, a stochastic policy c_n is always considered jointly with the design $\delta_{2,n} = \phi_n(D_{1,n}, U)$ used to generate the second wave data. Probabilities and expectations are thus taken with respect to the law induced by the rule ϕ_n generating the design $\delta_{2,n}$ given in the statements of Theorems 2 and 3. Probabilities and expectations in the limit experiment are likewise taken over the distribution of a random policy c induced by the associated limiting design δ_2 .

We first establish some properties of the quadratic program in the limit decision problem. Define the minimizer map

$$\Gamma(b) = \operatorname{argmin}_{c \in \mathcal{G}} r(c, b).$$

Lemma 4: *Under Assumption 6, $\Gamma(b)$ exists and is unique for all $b \in \mathbb{R}^k$.*

1. *For any $b_1, b_2 \in \mathbb{R}^k$, $\|\Gamma(b_1) - \Gamma(b_2)\| \leq \kappa^{-1} \|b_1 - b_2\|$*
2. *V is continuously differentiable with $\nabla V(b) = \Gamma(b)$*
3. *For any $b_1, b_2 \in \mathbb{R}^k$ we have $0 \leq r(\Gamma(b_1), b_2) - r(\Gamma(b_2), b_2) \leq \kappa^{-1} \|b_1 - b_2\|^2$*

Proof. Existence and uniqueness follow from the fact that $\mathcal{G}$ is a closed convex cone and $r(\cdot, b)$ is strongly convex for all b . Optimality of $\Gamma(b_1)$ implies that $(\bar{H}\Gamma(b_1) +$

$b_1)'(\Gamma(b_2) - \Gamma(b_1)) \geq 0$. Add this to the corresponding inequality for $\Gamma(b_2)$ and use strong convexity to obtain

$$\kappa \|\Gamma(b_1) - \Gamma(b_2)\|^2 \leq (b_1 - b_2)'(\Gamma(b_2) - \Gamma(b_1)) \leq \|b_1 - b_2\| \|\Gamma(b_1) - \Gamma(b_2)\|$$

which proves item 1. Danskin's theorem implies that $\nabla V(b) = \Gamma(b)$ and continuity follows from item 1, thus proving item 2. For item 3, the first inequality is implied by optimality of $\Gamma(b_2)$ and the second is because

$$\begin{aligned} r(\Gamma(b_1), b_2) - r(\Gamma(b_2), b_2) &= r(\Gamma(b_1), b_1) - r(\Gamma(b_2), b_1) + (b_2 - b_1)'(\Gamma(b_1) - \Gamma(b_2)) \\ &\leq (b_2 - b_1)'(\Gamma(b_1) - \Gamma(b_2)) \\ &\leq \kappa^{-1} \|b_1 - b_2\|^2 \end{aligned}$$

where the inequalities are from (i) optimality of $\Gamma(b_1)$ and (ii) item 1. $\square$

For θ in a neighborhood of θ_0 , let

$$\pi^*(\theta) \in \arg \max_{\pi \in \Pi: g(\pi) \leq 0} W(\pi, \theta)$$

be an oracle policy that the decision maker would like to choose if θ were known. We decompose regret into the smooth and oracle components

$$\begin{aligned} R_n(c, h) &= R_n^s(c, h) + R_n^o(h) \\ R_n^s(c, h) &= n[W(\pi_0, \theta_0 + h/\sqrt{n}) - W(\pi_0 + c/\sqrt{n}, \theta_0 + h/\sqrt{n})] \\ R_n^o(h) &= n[W(\pi^*(\theta_0 + h/\sqrt{n}), \theta_0 + h/\sqrt{n}) - W(\pi_0, \theta_0 + h/\sqrt{n})]. \end{aligned}$$

Lemma 5: *Suppose Assumption 6 holds. If $m_n \rightarrow m$ and $c_n \rightarrow c$ are deterministic sequences such that $\pi_0 + c_n/\sqrt{n} \in \Pi$ for all n , then*

$$\liminf_{n \rightarrow \infty} R_n^s(c_n, m_n) \geq \begin{cases} r(c, Bm), & \text{if } c \in \mathcal{G}, \\ \infty, & \text{otherwise.} \end{cases}$$

Moreover, for every $c \in \mathcal{G}$ and $m \in \mathbb{R}^\ell$ there exists a deterministic sequence $c_n \rightarrow c$ such that $\pi_0 + c_n/\sqrt{n} \in \Pi$ for all n and $R_n^s(c_n, m_n) \rightarrow r(c, Bm)$ whenever $m_n \rightarrow m$.

Proof. For this proof, use $a_j = \nabla g_j(\pi_0)$ and $G_j = \nabla^2 g_j(\pi_0)$. Taylor expansions of

centered regret and an active constraint j yield

$$\begin{aligned} R_n^s(c_n, m_n) &= \sqrt{n}D'c_n + c'_n Bm_n + \frac{1}{2}c'_n Hc_n + o(1) \\ 0 &\geq n^{-1/2}a'_j c_n + n^{-1}\frac{1}{2}c'_n G_j c_n + o(n^{-1}). \end{aligned}$$

The latter implies that $a'_j c \leq 0$ for all active constraints j . Stationarity implies $D = -\sum_{j \in \mathcal{J}_s} \lambda_{0j} a_j$. Thus, if $a'_j c < 0$ for some strictly active constraint $j \in \mathcal{J}_s$, then $R_n^s(c_n, m_n) \rightarrow \infty$ and the result holds. On the other hand, if $c \in \mathcal{G}$, then $a'_j c = 0$ for all $j \in \mathcal{J}_s$ and substituting for D in the Taylor expansion of R_n^s yields

$$\begin{aligned} R_n^s(c_n, m_n) &= -\sqrt{n} \sum_{j \in \mathcal{J}_s} \lambda_{0j} a'_j c_n + c'_n Bm_n + \frac{1}{2}c'_n Hc_n + o(1) \\ &\geq c'_n Bm_n + \frac{1}{2}c'_n Hc_n + \frac{1}{2} \sum_{j \in \mathcal{J}_s} \lambda_{0j} c'_n G_j c_n + o(1) \end{aligned}$$

and hence $\liminf_{n \rightarrow \infty} R_n^s(c_n, m_n) \geq r(c, Bm)$ as desired.

We now turn to constructing the recovery sequence. Fix $c \in \mathcal{G}$ and define $\mathcal{J}_c = \{j \in \mathcal{J}_w \cup \mathcal{J}_s : a'_j c = 0\}$. Let $\nu_n \downarrow 0$ be a sequence dominating the Taylor remainder in the expansion of g_j for all $j \in \mathcal{J}_w \cup \mathcal{J}_s$. Linear independence of the constraints implies that there exists a right inverse of the gradient matrix $A = [a'_j]_{j \in \mathcal{J}_c}$, which we can use to solve for a vector $\tilde{c}_n$ such that $a'_j \tilde{c}_n = -\frac{1}{2}c' G_j c - \nu_n$ for all $j \in \mathcal{J}_s$ and $a'_j \tilde{c}_n = -\frac{1}{2}c' G_j c - 1$ for all $j \in \mathcal{J}_c \cap \mathcal{J}_w$. Then the sequence

$$c_n = c + n^{-1/2} \tilde{c}_n$$

is feasible for all sufficiently large n . Substituting c_n into the Taylor expansion of R_n^s and taking limits yields $R_n^s(c_n, m_n) \rightarrow r(c, Bm)$ as desired. $\square$

Lemma 6: *Suppose Assumption 6 holds. For every compact set $M \subset \mathbb{R}^\ell$,*

$$\begin{aligned} \sup_{m \in M} \|\sqrt{n}(\pi^*(\theta_0 + m/\sqrt{n}) - \pi_0) - \Gamma(Bm)\| &\rightarrow 0 \\ \sup_{m \in M} |R_n^o(m) + V(Bm)| &\rightarrow 0 \end{aligned}$$

Proof. Assumption 6 implies there exists an $a > 0$ and a neighborhood N_π of π_0 such

that the quadratic growth condition holds:

$$W(\pi_0, \theta_0) - W(\pi, \theta_0) \geq a\|\pi - \pi_0\|^2$$

for all $\pi \in \Pi \cap N_\pi$. The compact level set condition, continuity of W , and uniqueness of π_0 imply that any sequence of feasible optimizers $\pi^*(\theta)$ for $\theta \rightarrow \theta_0$ eventually lies in N_π . Continuity of $\nabla_{\pi, \pi}^2 W$ and positive definiteness of H imply that $W(\cdot, \theta)$ is strongly concave on $N_\pi \cap \Pi$ and hence $\pi^*(\theta)$ is unique for all θ sufficiently close to θ_0 . For such a sequence $\pi^*(\theta)$, the previous display implies

$$a\|\pi^*(\theta) - \pi_0\|^2 \leq [W(\pi^*(\theta), \theta) - W(\pi^*(\theta), \theta_0)] - [W(\pi_0, \theta) - W(\pi_0, \theta_0)].$$

By the mean value theorem and continuity of $\nabla^2 W$, the right side is bounded by $C\|\theta - \theta_0\|\|\pi^*(\theta) - \pi_0\|$ for some constant C . Divide both sides by $\|\pi^*(\theta) - \pi_0\|$ (the case of equality is trivial) to obtain

$$\|\pi^*(\theta) - \pi_0\| \leq a^{-1}C\|\theta - \theta_0\|$$

which is bounded by $a^{-1}C \sup_{m \in M} \|m\|/\sqrt{n}$ for $\theta = \theta_0 + m/\sqrt{n}$ and sufficiently large n .

Assume to the contrary that the optimizer convergence result does not hold. Then there exists $n_j, m_j \in M$, and feasible optimizers $\pi_j = \pi^*(\theta_0 + m_j/\sqrt{n_j})$ such that the result does not hold. Pass to a subsequence along which $m_j \rightarrow m$ for some $m \in M$. By the previous paragraph, $c_j = \sqrt{n_j}(\pi_j - \pi_0)$ is bounded, and so there is a further convergent subsequence $c_j \rightarrow c$. By Lemma 5, $\liminf_{j \rightarrow \infty} R_{n_j}^s(c_j, m_j) \geq r(c, Bm)$.

Also by Lemma 5, for any $\tilde{c} \in \mathcal{G}$, there is a sequence $\tilde{c}_j \rightarrow \tilde{c}$ which is feasible for all j and $R_{n_j}^s(\tilde{c}_j, m_j) \rightarrow r(\tilde{c}, Bm)$. Optimality of π_j then implies

$$R_{n_j}^s(c_j, m_j) \leq R_{n_j}^s(\tilde{c}_j, m_j)$$

and therefore $r(c, Bm) \leq r(\tilde{c}, Bm)$ for all $\tilde{c} \in \mathcal{G}$. By Lemma 4, $c = \Gamma(Bm)$ and $\|c_j - \Gamma(Bm_j)\| \rightarrow 0$, a contradiction.

Finally, we establish the uniform convergence of the value function. Take the $\tilde{c}$ from the previous paragraph to be $\Gamma(Bm)$. Then $R_{n_j}^s(\tilde{c}_j, m_j) \rightarrow r(\Gamma(Bm), Bm) = V(Bm)$. Optimality of π_j implies $\limsup_{j \rightarrow \infty} R_{n_j}^s(c_j, m_j) \leq V(Bm)$, and so $\lim_{j \rightarrow \infty} R_{n_j}^s(c_j, m_j) =$

$V(Bm)$. Uniformity follows from the same contradiction argument as in the previous paragraph. $\square$

Corollary 2: *Suppose Assumption 6 holds. For every $K < \infty$ and data $(D_{1,n}, D_{2,n})$, there exists $n_K < \infty$ and $C_K < \infty$ such that for all $n \geq n_K$, the optimal policy $\pi_{n,K}^*$ exists and is unique, and $\sqrt{n}\|\pi_{n,K}^* - \pi_0\| \leq C_K$.*

Proof. For n large enough, all parameters $\theta = \theta_0 + h/\sqrt{n}$ with h in the support of the posterior lie in a neighborhood N_θ on which the level set condition, twice continuous differentiability, and strong concavity of W hold. The uniqueness argument in the proof of Lemma 6 applies uniformly over N , so $\pi_{n,K}^*$ exists and is unique. Optimality of $\pi_{n,K}^*$ implies that

$$\begin{aligned} a\|\pi_{n,K}^* - \pi_0\|^2 &\leq \mathbb{E}_K \left[W\left(\pi_{n,K}^*, \theta_0 + \frac{h}{\sqrt{n}}\right) - W(\pi_{n,K}^*, \theta_0) \right. \\ &\quad \left. - W\left(\pi_0, \theta_0 + \frac{h}{\sqrt{n}}\right) + W(\pi_0, \theta_0) \mid D_{1,n}, D_{2,n} \right] \\ &\leq \frac{CK}{\sqrt{n}} \|\pi_{n,K}^* - \pi_0\| \end{aligned}$$

which establishes the desired bound. $\square$

Proof of Theorem 2. Since $\mathcal{R}_{n,K}(\delta_{2,n}, c_n) \geq \mathcal{R}_{n,K}^*(\delta_{2,n})$ for any c_n , it suffices to restrict attention to the optimal policy $c_{n,K}^* = \sqrt{n}(\pi_{n,K}^* - \pi_0)$.

We first take a subsequence along which $\mathcal{R}_{n,K}^*(\delta_{2,n}) \rightarrow \liminf_{n \rightarrow \infty} \mathcal{R}_{n,K}^*(\delta_{2,n})$. If the lim inf is infinite, the result holds trivially, and so we proceed under the assumption that it is finite. Since $c_{n,K}^*$ is tight by Corollary 2, Theorem 1 and Corollary 1 imply that every subsequence has a further subsequence along which $(\delta_{2,n}, c_{n,K}^*) \xrightarrow{h} (\delta_2, c)$ for any h , where (δ_2, c) is a design/policy in the limit experiment.

Since the sequence $c_{n,K}^*$ converges in distribution to c along the subsequence for every h , we can extract a Skorokhod representation for each h such that $c_{n,K}^* \rightarrow c$ almost surely. Lemma 5 implies that $\liminf_{n \rightarrow \infty} R_n^s(c_{n,K}^*, h) \geq r(c, Bh)$ for every h if $c \in \mathcal{G}$ and $\liminf_{n \rightarrow \infty} R_n^s(c_{n,K}^*, h) = \infty$ otherwise. Lemma 6 implies that $R_n^o(h) \rightarrow -V(Bh)$ for every h . Thus, $\liminf_{n \rightarrow \infty} R_n(c_{n,K}^*, h) \geq R_\infty(c, Bh)$ for every h .

Thus, by Fatou's Lemma and Assumption 7,

$$\liminf_{n \rightarrow \infty} \mathcal{R}_{n,K}^*(\delta_{2,n}) = \liminf_{n \rightarrow \infty} \int \mathbb{E}_h[R_n(c_{n,K}^*, h)] q_{n,K}(h) dh$$

$$\begin{aligned}
&\geq \int \liminf_{n \rightarrow \infty} \mathbb{E}_h [R_n(c_{n,K}^*, h)] q_{\infty,K}(h) dh \\
&\geq \int \mathbb{E}_h [R_{\infty}(c, Bh)] q_{\infty,K}(h) dh
\end{aligned}$$

and hence $\liminf_{n \rightarrow \infty} \mathcal{R}_{n,K}^*(\delta_{2,n}) \geq \mathcal{R}_{\infty,K}^{**}$. $\square$

C PROOF OF THEOREM 3

C.1 Fixed- K upper bound

Define the local first-wave estimators $\hat{h}_{1,n} = \sqrt{n}(\hat{\theta}_{1,n} - \theta_0)$ and $\hat{c}_{1,n} = \sqrt{n}(\hat{\pi}_{1,n} - \pi_0)$. We also define

$$\hat{S}_n(c, b) = \sqrt{n}(c - \hat{c}_{1,n})' \hat{D}_n + (c - \hat{c}_{1,n})'(b - \hat{B}_n \hat{h}_{1,n}) + \frac{1}{2}(c - \hat{c}_{1,n})' \hat{H}_n (c - \hat{c}_{1,n})$$

and

$$\hat{\Gamma}_n(b) = \operatorname{argmin}_c \hat{S}_n(c, b) \quad \text{s.t.} \quad g(\pi_0 + c/\sqrt{n}) \leq 0$$

and note that $\hat{c}_n^{\text{os}} = \hat{\Gamma}_n(\hat{B}_n \hat{h}_n^{\text{os}})$. We also define $\hat{\beta}_n(m) = \hat{\beta}_{1,n} + \frac{1}{\sqrt{n}} \hat{B}_n(m - \hat{h}_{1,n})$.

We use $\mathbb{E}_K$ to denote expectations with respect to the K -truncated prior. $\mathbb{E}_{\infty}$ is used to denote expectations with respect to the flat prior. Let $\xi \sim N(0, I)$ independently of the first-wave data. The simulated posterior mean of h is given by

$$\tilde{h}_n(d, \xi) = \hat{h}_{1,n} + \left(\frac{n}{n_1} \hat{J}_{1,n}^{-1} - \hat{J}_n(d)^{-1} \right)^{1/2} \xi$$

and the finite-sample loss minimized by $\hat{\delta}_{2,n}$ is

$$\hat{L}_n(d) = \mathbb{E}_{\xi} \left[n \hat{V}_n(\hat{\beta}_n(\tilde{h}_n(d, \xi))) \right].$$

The expectation $\mathbb{E}_{\xi}$ integrates only over the randomness in ξ and not the first-wave data. The posterior mean of h in the limit experiment after the first wave is

$$\hat{h}_{1,\infty} = (1 - \rho)^{-1} J_1^{-1} A_1$$

and the flat-prior loss in the limit experiment is

$$L_\infty(d) = \mathbb{E}_\xi \left[V(B(\hat{h}_{1,\infty} + ((1 - \rho)^{-1} J_1^{-1} - J(d)^{-1})^{1/2} \xi)) \right]$$

Define the K -truncated loss in the limit experiment as

$$L_\infty^K(d) = \mathbb{E}_K \left[V(\hat{b}_{2,K}(d)) \mid A_1, U \right].$$

We also define the centered loss criterion

$$\hat{L}_n^c(d) = \hat{L}_n(d) - \hat{L}_n(d_0) \quad L_\infty^c(d) = L_\infty(d) - L_\infty(d_0)$$

for a fixed $d_0 \in \Delta$. This centering does not affect the minimizer. We make the dependence of $\hat{h}_n^{\text{os}}$ on the design $d \in \Delta$ explicit by writing it as a process $\hat{h}_n^{\text{os}}(d)$ where necessary. The limit analog is

$$\hat{h}^{\text{os}}(d) = J(d)^{-1}(A_1 + A_2(d))$$

and the corresponding flat-prior policy is

$$c_\infty^{\text{flat}} = \Gamma(B\hat{h}^{\text{os}}(d)).$$

Lemma 7: *Suppose Assumptions 6 and 8 hold. Then*

- *for every compact set $M \subset \mathbb{R}^\ell$, $\sup_{m \in M} \|\hat{\Gamma}_n(\hat{B}_n m) - \hat{c}_{1,n}\| = O_P(1)$*
- *With probability approaching one, the optimization problem defining $\hat{\Gamma}_n$ has a unique solution, and $0 \leq -n\hat{V}_n(\hat{\beta}_n(m)) \leq \kappa^{-1}\|\hat{B}_n\|^2\|m - \hat{h}_{1,n}\|^2$ for every $m \in \mathbb{R}^\ell$.*

Proof. Lemma 6, consistency of first wave estimates, and continuous differentiability of W imply that $\hat{H}_n \xrightarrow{P} H$ and $\hat{B}_n \xrightarrow{P} B$. Thus, Assumption 6 implies that with probability approaching one, $u'\hat{H}_n u \geq \frac{\kappa}{2}\|u\|^2$ for all $u \in \text{span}(\Pi - \Pi)$. Convexity of Π and optimality of $\hat{\pi}_{1,n}$ then imply uniqueness of $\hat{\Gamma}_n(b)$. The argument in the proof of Lemma 4 then delivers both results. $\square$

To compare the estimated and limiting value functions, we define the centered value $D_V(b_1, b_2) = V(b_2) - V(b_1) - \Gamma(b_1)'(b_2 - b_1)$.

Lemma 8: Suppose Assumptions 1 to 8 hold. Then for every fixed h and every compact $M \subset \mathbb{R}^\ell$,

1. $\sup_{m \in M} |n\hat{V}_n(\hat{\beta}_n(m)) - D_V(B\hat{h}_{1,n}, Bm)| = o_{P_h}(1)$
2. $\sup_{m \in M} \|\hat{\Gamma}_n(\hat{B}_n m) - \Gamma(Bm)\| = o_{P_h}(1)$
3. $\sup_{m \in M} |R_n^s(\hat{\Gamma}_n(\hat{B}_n m), m) - V(Bm)| = o_{P_h}(1)$

Proof. Since $\hat{c}_{1,n}$ is tight by Lemma 6, a Taylor expansion of $R_n^s(\cdot, m)$ around $(\hat{\pi}_{1,n}, \hat{\theta}_{1,n})$ gives, for every compact $C \subseteq \mathbb{R}^k$,

$$\sup_{c \in C} \sup_{m \in M} |\hat{S}_n(c, \hat{B}_n m) - R_n^s(c, m) + R_n^s(\hat{c}_{1,n}, m)| = o_{P_h}(1).$$

Minimizing each term over c (justified by the localization of Lemmas 6 and 7) and using Lemma 6 to substitute $\min_c R_n^s(c, m) = V(Bm) + o_{P_h}(1)$ yields

$$\sup_{m \in M} |n\hat{V}_n(\hat{\beta}_n(m)) - V(Bm) + R_n^s(\hat{c}_{1,n}, m)| = o_{P_h}(1).$$

Twice continuous differentiability of W implies that

$$\sup_{m \in M} |R_n^s(\hat{c}_{1,n}, m) - R_n^s(\hat{c}_{1,n}, \hat{h}_{1,n}) - \hat{c}_{1,n}' B(m - \hat{h}_{1,n})| \rightarrow 0.$$

Lemma 6 gives $R_n^s(\hat{c}_{1,n}, \hat{h}_{1,n}) = V(B\hat{h}_{1,n}) + o_{P_h}(1)$. Lemma 6 also gives $\hat{c}_{1,n} = \Gamma(B\hat{h}_{1,n}) + o_{P_h}(1)$. Thus,

$$\sup_{m \in M} |n\hat{V}_n(\hat{\beta}_n(m)) - V(Bm) + V(B\hat{h}_{1,n}) + \Gamma(B\hat{h}_{1,n})' B(m - \hat{h}_{1,n})| = o_{P_h}(1)$$

which is item 1 of the lemma.

For item 2, suppose the conclusion fails. Then there exists a subsequence and m_n along which $\|\hat{\Gamma}_n(\hat{B}_n m_n) - \Gamma(Bm_n)\| \geq \epsilon$ with nonvanishing probability. By Lemma 7, $\hat{\Gamma}_n(\hat{B}_n m_n)$ is tight, and therefore has a weakly convergent subsequence. On a Skorokhod representation, $(m_n, \hat{\Gamma}_n(\hat{B}_n m_n)) \rightarrow (m, c)$ almost surely along the subsequence. Then Lemma 5 implies that there exists a feasible sequence $c_n \rightarrow \Gamma(Bm)$ almost surely such that

$$\hat{S}_n(\hat{\Gamma}_n(\hat{B}_n m_n), \hat{B}_n m_n) \leq \hat{S}_n(c_n, \hat{B}_n m_n) \implies r(c, Bm) \leq r(\Gamma(Bm), Bm)$$

which, by uniqueness of the minimizer, implies $\hat{\Gamma}_n(\hat{B}_n m_n) \rightarrow \Gamma(Bm)$ in probability, a contradiction.

For item 3, note that the proof of item 1 implies that

$$\begin{aligned} R_n^s(\hat{\Gamma}_n(\hat{B}_n m), m) &= D_V(B\hat{h}_{1,n}, Bm) + V(B\hat{h}_{1,n}) + \Gamma(B\hat{h}_{1,n})'B(m - \hat{h}_{1,n}) + o_{P_h}(1) \\ &= V(Bm) + o_{P_h}(1) \end{aligned}$$

uniformly in $m \in M$. □

Lemma 9: *Suppose Assumptions 1 to 8 hold. Then*

1. $\sup_{d \in \Delta} \|\hat{h}_n^{os}(d) - J(d)^{-1}(A_{1,n} + A_{2,n}(d))\| = o_{P_0}(1)$ and $\hat{h}_n^{os}(\cdot) \xrightarrow{0} \hat{h}^{os}(\cdot)$ in $\ell^\infty(\Delta)$
2. $\hat{L}_n^c(\cdot) \xrightarrow{h} L_\infty^c(\cdot)$ in $\ell^\infty(\Delta)$ for any h .

Proof. Since $A_{2,n}(\cdot)$ converges to a Gaussian process, its covariance kernel is asymptotically equicontinuous. Assumptions 1, 5, and compactness of Δ imply that $\sup_{d \in \Delta} \|\hat{J}_n(d) - J(d)\| = o_{P_0}(1)$. Stochastic equicontinuity of $A_{2,n}(\cdot)$, uniform consistency of $\hat{J}_n(\cdot)$, uniform nonsingularity of $J(\cdot)$, and the continuous mapping theorem then imply the first result.

The first-wave score equation and Assumption 8 imply

$$\hat{h}_{1,n} = ((1 - \rho)J_1)^{-1}A_{1,n} + o_{P_0}(1).$$

Then the second-wave score expansion of Assumption 8 gives the first result of item

1. Lemma 3 then gives the second result of item 1.

Lemma 3 and contiguity of the first wave also give

$$\hat{h}_{1,n} \xrightarrow{h} \hat{h}_{1,\infty} \quad \hat{J}_{1,n}, \hat{J}_n(\cdot) \xrightarrow{P_h} J_1, J(\cdot).$$

Tightness of $\hat{h}_{1,n}$ and uniform boundedness in probability of $\frac{n}{n_1}\hat{J}_{1,n}^{-1} - \hat{J}_n(\cdot)^{-1}$ imply that $\tilde{h}_n(d, \xi)$ is tight uniformly in $d \in \Delta$. Lemma 8 then implies that

$$\sup_{d \in \Delta} \left| n\hat{V}_n(\hat{\beta}_n(\tilde{h}_n(d, \xi))) - D_V(B\hat{h}_{1,n}, B\tilde{h}_n(d, \xi)) \right| = o_{P_h}(1).$$

Lemma 7 gives

$$|n\hat{V}_n(\hat{\beta}_n(\tilde{h}_n(d, \xi))) - D_V(B\hat{h}_{1,n}, B\tilde{h}_n(d, \xi))| \leq O_{P_h}(1)\|\xi\|^2$$

which establishes uniform integrability of the left-hand side. This implies that

$$\sup_{d \in \Delta} \mathbb{E}_\xi \left| n \hat{V}_n(\hat{\beta}_n(\tilde{h}_n(d, \xi))) - D_V(B\hat{h}_{1,n}, B\tilde{h}_n(d, \xi)) \right| = o_{P_n}(1)$$

and therefore

$$\sup_{d \in \Delta} |\hat{L}_n(d) - \mathbb{E}_\xi[D_V(B\hat{h}_{1,n}, B\tilde{h}_n(d, \xi))]| = o_{P_n}(1).$$

The continuous mapping theorem then implies that

$$\begin{aligned} \mathbb{E}_\xi[D_V(B\hat{h}_{1,n}, B\tilde{h}_n(\cdot, \xi))] &\overset{h}{\rightsquigarrow} \mathbb{E}_\xi[D_V(B\hat{h}_{1,\infty}, B(\hat{h}_{1,\infty} + ((1-\rho)^{-1}J_1^{-1} - J(\cdot)^{-1})^{1/2}\xi))] \\ &= L_\infty(\cdot) - V(B\hat{h}_{1,\infty}) \end{aligned}$$

in $\ell^\infty(\Delta)$. Applying the continuous map $f \rightarrow f - f(d_0)$ yields $\hat{L}_n^c(\cdot) \overset{h}{\rightsquigarrow} L_\infty^c(\cdot)$ in $\ell^\infty(\Delta)$. $\square$

Proposition 1: Fix $K < \infty$. Suppose Assumptions 1 to 8 hold. Then there exists some $\delta_{2,\infty,K}^{\text{flat}}$ in the limit experiment such that $\delta_{2,\infty,K}^{\text{flat}}$ minimizes $L_\infty(\cdot)$ and

$$\limsup_{n \rightarrow \infty} \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}) \leq \mathcal{R}_{\infty,K}(\delta_{2,\infty,K}^{\text{flat}}, c_\infty^{\text{flat}})$$

Proof. Pass to a subsequence for which the limsup is attained. By Lemmas 8 and 9, $\hat{c}_n^{\text{os}}$ is tight. Corollary 1 then implies that there exists a further subsequence and $\delta_{2,\infty,K}^{\text{flat}}$ in the limit experiment such that for every h ,

$$\left(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}, A_{1,n}, A_{2,n}(\hat{\delta}_{2,n}) \right) \overset{h}{\rightsquigarrow} \left(\delta_{2,\infty,K}^{\text{flat}}, c_\infty, A_1, A_2(\delta_{2,\infty,K}^{\text{flat}}) \right).$$

First, we show that $\delta_{2,\infty,K}^{\text{flat}}$ minimizes $L_\infty(\cdot)$. Define $f(l, d) = l(d) - \inf_{d \in \Delta} l(d)$. This is a continuous function of (l, d) in $C(\Delta) \times \Delta$, and the limit process has almost surely continuous sample paths. Item 2 of Lemma 9, along with joint convergence of $\hat{\delta}_{2,n}$ and $A_{1,n}$, implies that

$$\left(\hat{\delta}_{2,n}, \hat{L}_n^c(\cdot) \right) \overset{h}{\rightsquigarrow} \left(\delta_{2,\infty,K}^{\text{flat}}, L_\infty^c(\cdot) \right).$$

Since $\hat{\delta}_{2,n}$ minimizes $\hat{L}_n^c(\cdot)$, the extended continuous mapping theorem implies that $f(L_\infty^c, \delta_{2,\infty,K}^{\text{flat}}) = 0$, and so $\delta_{2,\infty,K}^{\text{flat}}$ minimizes $L_\infty(\cdot)$.

Item 1 of Lemma 9 implies that

$$\hat{h}_n^{\text{os}}(\hat{\delta}_{2,n}) - J(\hat{\delta}_{2,n})^{-1}(A_{1,n} + A_{2,n}(\hat{\delta}_{2,n})) = o_{P_0}(1)$$

Contiguity as established in Lemma 3 then implies that this is $o_{P_h}(1)$ for every h . Thus,

$$\hat{h}_n^{\text{os}}(\hat{\delta}_{2,n}) \xrightarrow{h} J(\delta_{2,\infty,K}^{\text{flat}})^{-1}(A_1 + A_2(\delta_{2,\infty,K}^{\text{flat}})) = \hat{h}^{\text{os}}(\delta_{2,\infty,K}^{\text{flat}}).$$

Items 2 and 3 of Lemma 8 imply that

$$\begin{aligned} \|\hat{c}_n^{\text{os}} - \Gamma(B\hat{h}_n^{\text{os}}(\hat{\delta}_{2,n}))\| &= o_{P_h}(1) \\ R_n^s(\hat{c}_n^{\text{os}}, \hat{h}_n^{\text{os}}(\hat{\delta}_{2,n})) &= V(B\hat{h}_n^{\text{os}}(\hat{\delta}_{2,n})) + o_{P_h}(1) \end{aligned}$$

which implies that $(\hat{h}_n^{\text{os}}(\hat{\delta}_{2,n}), \hat{c}_n^{\text{os}}) \xrightarrow{h} (\hat{h}^{\text{os}}(\delta_{2,\infty,K}^{\text{flat}}), \Gamma(B\hat{h}^{\text{os}}(\delta_{2,\infty,K}^{\text{flat}})))$ which identifies the limiting policy as $c_\infty = c_\infty^{\text{flat}} = \Gamma(B\hat{h}^{\text{os}}(\delta_{2,\infty,K}^{\text{flat}}))$ almost surely.

Twice continuous differentiability of W implies that

$$\sup_{c \in C} \sup_{m_1, m_2 \in M} |R_n^s(c, m_1) - R_n^s(c, m_2) - c'B(m_1 - m_2)| \rightarrow 0$$

for every compact C and M . Since $(\hat{c}_n^{\text{os}}, \hat{h}_n^{\text{os}}(\hat{\delta}_{2,n}))$ is tight, this implies that

$$\begin{aligned} R_n^s(\hat{c}_n^{\text{os}}, h) &= R_n^s(\hat{c}_n^{\text{os}}, \hat{h}_n^{\text{os}}(\hat{\delta}_{2,n})) + \hat{c}_n^{\text{os}'} B(h - \hat{h}_n^{\text{os}}(\hat{\delta}_{2,n})) + o_{P_h}(1) \\ &= V(B\hat{h}_n^{\text{os}}(\hat{\delta}_{2,n})) + \hat{c}_n^{\text{os}'} B(h - \hat{h}_n^{\text{os}}(\hat{\delta}_{2,n})) + o_{P_h}(1) \\ &\xrightarrow{h} r(c_\infty^{\text{flat}}, Bh) \end{aligned}$$

by the continuous mapping theorem. Since $R_n^o(h) \rightarrow -V(Bh)$ by Lemma 6, we have $R_n(\hat{c}_n^{\text{os}}, h) \xrightarrow{h} R_\infty(c_\infty^{\text{flat}}, Bh)$ for every h . The uniform integrability condition of Assumption 8 then implies that

$$\lim_{n \rightarrow \infty} \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}) = \mathcal{R}_{\infty,K}(\delta_{2,\infty,K}^{\text{flat}}, c_\infty^{\text{flat}})$$

along our subsequence, establishing the lim sup along the full sequence. $\square$

C.2 Taking the large- K limit

It remains to close the gap between the upper bound given in Proposition 1 and the lower bound given in Theorem 2. The next lemma collects some useful properties of the limiting problem. Define

$$\hat{b}_{1,K} = \mathbb{E}_K[Bh \mid A_1], \quad \hat{b}_{1,\infty} = B\hat{h}_{1,\infty}, \quad \hat{b}_{2,\infty}(d) = B\hat{h}^{\text{os}}(d).$$

Lemma 10: *Suppose Assumption 5 holds. Then for any sequence of designs δ_K in the limit experiment,*

$$\begin{aligned} \mathbb{E}_K \left[\|\hat{b}_{1,K} - \hat{b}_{1,\infty}\|^2 \right] &\rightarrow 0 \\ \mathbb{E}_K \left[\|\hat{b}_{2,K}(\delta_K) - \hat{b}_{2,\infty}(\delta_K)\|^2 \right] &\rightarrow 0. \end{aligned}$$

Proof. We prove the second result; the first is similar. Let $\hat{h}_{2,\infty,K}$ be the mean of the posterior distribution on h restricted to the K -ball $\mathcal{B}_K$. Fix $M > 0$. On the event where $\text{dist}(h, \mathcal{B}_K^c) > 2M$ and $\|\hat{h}_{2,\infty,\infty} - h\| \leq M$, we have $\text{dist}(\hat{h}_{2,\infty,K}, \mathcal{B}_K^c) \geq M$. Thus,

$$\|\hat{h}_{2,\infty,K} - \hat{h}_{2,\infty,\infty}\| \leq \frac{\mathbb{E}[\|\xi\| \mathbb{1}[\|\xi\| \geq M]]}{1 - P(\|\xi\| \geq M)}$$

for a $N(0, J(d)^{-1})$ random vector ξ . The right-hand side tends to zero uniformly over $d \in \Delta$ as $M \rightarrow \infty$.

On the other hand, the event where $\text{dist}(h, \mathcal{B}_K^c) \leq 2M$ has mass $1 - (1 - 2M/K)^\ell \leq 2\ell M/K$. The inequality $\mathbb{E}_K \|\hat{h}_{2,\infty,K} - \hat{h}_{2,\infty,\infty}\|^4 \leq \mathbb{E}_K \|\hat{h}_{2,\infty,\infty} - h\|^4$ establishes a uniform bound on fourth moments over designs, and hence uniform integrability, so that $\mathbb{E}_K \|\hat{h}_{2,\infty,K} - \hat{h}_{2,\infty,\infty}\|^2 \rightarrow 0$. The result then follows by multiplying by B . $\square$

Lemma 11: *Suppose Assumptions 5 and 6 hold. Then for every sequence of designs δ_K in the limit experiment,*

$$\mathbb{E}_K \left| L_\infty^K(\delta_K) - V(\hat{b}_{1,K}) - L_\infty(\delta_K) + V(\hat{b}_{1,\infty}) \right| \rightarrow 0.$$

Proof. We begin by showing that the K -truncated and flat posteriors in the limit experiment converge in total variation as K grows. Let $Q_{1,K} \propto N(\hat{h}_{1,\infty}, (1-\rho)^{-1}J_1^{-1})\mathbb{1}[\|h\| \leq K]$ and $Q_{1,\infty} = N(\hat{h}_{1,\infty}, (1-\rho)^{-1}J_1^{-1})$ be the first wave posteriors in the limit

experiment. Then $\|Q_{1,K} - Q_{1,\infty}\|_{\text{TV}} = Q_{1,\infty}(\|h\| > K)$. On the event where $\text{dist}(h, \mathcal{B}_K^c) > 2M$ and $\|\hat{h}_{1,\infty} - h\| \leq M$, we have $Q_{1,\infty}(\|h\| > K) \leq P(\|\xi\| \geq M)$. The probability of the complement of this event is bounded by $P(\text{dist}(h, \mathcal{B}_K^c) \leq 2M) + P(\|\hat{h}_{1,\infty} - h\| > M)$. Thus,

$$\begin{aligned} \mathbb{E}_K \|Q_{1,K} - Q_{1,\infty}\|_{\text{TV}} &\leq P(\|\xi\| \geq M) + P(\text{dist}(h, \mathcal{B}_K^c) \leq 2M) + P(\|\hat{h}_{1,\infty} - h\| > M) \\ &\leq P(\|\xi\| \geq M) + 2\ell M/K + P(\|\hat{h}_{1,\infty} - h\| > M) \end{aligned}$$

which tends to zero letting $K \rightarrow \infty$ and then $M \rightarrow \infty$.

We now show that the posterior predictive distributions of the second wave data converge in total variation. Conditional on (h, A_1, U) , the observation A_2 follows the same Gaussian law in the K -truncated and flat limit experiments. Since total variation contracts under a Markov kernel, we have

$$\left\| \int \Phi(\delta_2) dQ_{1,K} - \int \Phi(\delta_2) dQ_{1,\infty} \right\|_{\text{TV}} \leq \|Q_{1,K} - Q_{1,\infty}\|_{\text{TV}}$$

where $\Phi(d)$ is the Gaussian kernel $h \mapsto N(\rho J_2(d)h, \rho J_2(d))$. Thus, the posterior predictive distributions of A_2 converge in total variation as well.

Finally, we show that the centered expected loss functions converge. The posterior means $\hat{h}_{2,\infty,K}$ and $\hat{h}_{2,\infty,\infty}$ are martingales under their predictive laws, so

$$\begin{aligned} L_\infty^K(\delta_K) - V(\hat{b}_{1,K}) &= \mathbb{E}_K[D_V(\hat{b}_{1,K}, \hat{b}_{2,K}(\delta_K)) \mid A_1, U] \\ L_\infty(\delta_K) - V(\hat{b}_{1,\infty}) &= \mathbb{E}_\infty[D_V(\hat{b}_{1,\infty}, \hat{b}_{2,\infty}(\delta_K)) \mid A_1, U]. \end{aligned}$$

By Lemma 4,

$$\begin{aligned} &|D_V(\hat{b}_{1,K}, \hat{b}_{2,K}(\delta_K)) - D_V(\hat{b}_{1,\infty}, \hat{b}_{2,\infty}(\delta_K))| \\ &\leq 2\kappa^{-1}(\|\hat{b}_{2,K}(\delta_K) - \hat{b}_{1,K}\| + \|\hat{b}_{2,\infty}(\delta_K) - \hat{b}_{1,\infty}\|) \times (\|\hat{b}_{1,K} - \hat{b}_{1,\infty}\| + \|\hat{b}_{2,K}(\delta_K) - \hat{b}_{2,\infty}(\delta_K)\|) \end{aligned}$$

Consequently,

$$\mathbb{E}_K[|D_V(\hat{b}_{1,K}, \hat{b}_{2,K}(\delta_K)) - D_V(\hat{b}_{1,\infty}, \hat{b}_{2,\infty}(\delta_K))| \mid A_1, U] \rightarrow 0$$

in L_1 by Lemma 10.

It remains to show that

$$\mathbb{E}_K[D_V(\hat{b}_{1,\infty}, \hat{b}_{2,\infty}(\delta_K)) \mid A_1, U] - \mathbb{E}_\infty[D_V(\hat{b}_{1,\infty}, \hat{b}_{2,\infty}(\delta_K)) \mid A_1, U]$$

converges to 0 in L_1 under the K -truncated prior. By concavity of V and Lemma 4,

$$-\frac{1}{2\kappa}\|b_2 - b_1\|^2 \leq D_V(b_1, b_2) \leq 0$$

which establishes uniform second moment bounds on D_V through fourth moment bounds on the Gaussian posterior increment $\hat{b}_{2,\infty}(\delta_K) - \hat{b}_{1,\infty}$. The inequality

$$\begin{aligned} & \left| \int D_V \Phi(\delta_2) Q_{1,K}(dh) - \int D_V \Phi(\delta_2) Q_{1,\infty}(dh) \right| \\ & \leq \left[2\|\Phi(\delta_2)Q_{1,K} - \Phi(\delta_2)Q_{1,\infty}\|_{\text{TV}} \left(\int D_V^2 \Phi(\delta_2) Q_{1,K} + \int D_V^2 \Phi(\delta_2) Q_{1,\infty} \right) \right]^{1/2} \end{aligned}$$

combined with the total variation convergence established above implies that the difference in conditional expectations converges to zero in L_1 under the K -truncated prior. $\square$

Lemma 12: Under Assumption 6, $\mathcal{R}_{\infty,K}^{**} < \infty$ for every K .

Proof. For any $c \in \mathcal{G}$, we have $R_\infty(c, Bh) = r(c, Bh) - V(Bh) \geq 0$. The feasible policy $c = 0$ has regret $R_\infty(0, Bh) = -V(Bh) \leq \frac{1}{2\kappa}\|Bh\|^2$ by Assumption 6. Since $q_{\infty,K}$ has compact support, it has finite second moments, and therefore $\mathcal{R}_{\infty,K}^{**} \leq \int \frac{1}{2\kappa}\|Bh\|^2 q_{\infty,K}(h)dh < \infty$. $\square$

Proposition 2: Suppose Assumptions 5 and 6 hold. Let $\delta_{2,\infty,K}^{\text{flat}}$ be a sequence of feasible designs, possibly depending on K , each of which minimizes $L_\infty(\cdot)$ in the limit experiment. Then

$$\lim_{K \rightarrow \infty} \left\{ \mathcal{R}_{\infty,K}(\delta_{2,\infty,K}^{\text{flat}}, c_\infty^{\text{flat}}) - \mathcal{R}_{\infty,K}^{**} \right\} = 0.$$

Proof. Choose an admissible design sequence $\delta_{2,\infty,K}$ satisfying $\mathcal{R}_{\infty,K}^*(\delta_{2,\infty,K}) \leq \mathcal{R}_{\infty,K}^{**} + \varepsilon_K$ for some $\varepsilon_K \rightarrow 0$. By this choice of $\delta_{2,\infty,K}$ and Lemma 11,

$$\begin{aligned} 0 & \leq \mathbb{E}_K[L_\infty^K(\delta_{2,\infty,K}^{\text{flat}}) - L_\infty^K(\delta_{2,\infty,K})] + \varepsilon_K \\ & \leq \mathbb{E}_K[L_\infty(\delta_{2,\infty,K}^{\text{flat}}) - L_\infty(\delta_{2,\infty,K})] + o(1) + \varepsilon_K \rightarrow 0 \end{aligned}$$

which shows that $0 \leq \mathcal{R}_{\infty,K}^*(\delta_{2,\infty,K}^{\text{flat}}) - \mathcal{R}_{\infty,K}^{**} \rightarrow 0$ as $K \rightarrow \infty$. Lemma 4 then implies that

$$\begin{aligned} 0 &\leq \mathcal{R}_{\infty,K}(\delta_{2,\infty,K}^{\text{flat}}, c_{\infty}^{\text{flat}}) - \mathcal{R}_{\infty,K}^*(\delta_{2,\infty,K}^{\text{flat}}) \\ &\leq \kappa^{-1} \mathbb{E}_K[\|\hat{b}_{2,\infty}(\delta_{2,\infty,K}^{\text{flat}}) - \hat{b}_{2,K}(\delta_{2,\infty,K}^{\text{flat}})\|^2] \end{aligned}$$

which converges to zero by Lemma 10. Combining the previous two displays gives the desired result. $\square$

We can now show that the proposed procedure is asymptotically optimal.

Proof of Theorem 3. By Theorem 2,

$$\limsup_{n \rightarrow \infty} \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}) \geq \liminf_{n \rightarrow \infty} \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}) \geq \mathcal{R}_{\infty,K}^{**}.$$

By Proposition 1,

$$\limsup_{n \rightarrow \infty} \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}) \leq \mathcal{R}_{\infty,K}(\delta_{2,\infty,K}^{\text{flat}}, c_{\infty}^{\text{flat}})$$

where the limiting design $\delta_{2,\infty,K}^{\text{flat}}$ may depend on K . Thus,

$$0 \leq \limsup_{n \rightarrow \infty} \mathcal{R}_{n,K}(\hat{\delta}_{2,n}, \hat{c}_n^{\text{os}}) - \mathcal{R}_{\infty,K}^{**} \leq \mathcal{R}_{\infty,K}(\delta_{2,\infty,K}^{\text{flat}}, c_{\infty}^{\text{flat}}) - \mathcal{R}_{\infty,K}^{**}$$

which converges to zero as $K \rightarrow \infty$ by Proposition 2. $\square$

D VERIFYING STOCHASTIC EQUICONTINUITY

We say a sequence of processes $A_n(\cdot)$ indexed by a compact Euclidean set Δ is stochastically equicontinuous under P if for every $\varepsilon > 0$ and $\eta > 0$ there exists an $\iota > 0$ such that

$$\limsup_{n \rightarrow \infty} P \left(\sup_{d, d' \in \Delta: \|d-d'\| < \iota} \|A_n(d) - A_n(d')\| > \eta \right) < \varepsilon. \quad (6)$$

Here we suppose the supremum is measurable.

For a scalar-valued class of functions $\mathcal{F} = \{f(\cdot, d) : d \in \Delta\}$ where $f(\cdot, d)$ is a function of the data w_i and the treatment assignment d , define the centered empirical

process

$$\mathbb{G}_n f(d) = \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} (f(w_i, d) - \mathbb{E}[f(w_i, d)]).$$

Note that we normalize by $\sqrt{n}$ but sum over only the second wave to match the processes in the main text.

Our primitive conditions for stochastic equicontinuity are based on the Type IV condition of Andrews (1994). This class is defined as follows. Let $p \geq 2$. Suppose there are constants $C_1, C_2, r_0 > 0$ such that

$$\sup_n \sup_{i \in \mathcal{I}_{2,n}} \left(E \sup_{\substack{\tilde{d} \in \Delta \\ \|d - \tilde{d}\| \leq r}} |f(w_i, d) - f(w_i, \tilde{d})|^p \right)^{1/p} \leq C_1 r^{C_2} \quad (7)$$

for every $d \in \Delta$ and $0 < r \leq r_0$. Then we say that $\mathcal{F}$ satisfies the Type IV condition with p .

We also will impose the following envelope condition. For a class of functions $\mathcal{F}$, the relevant envelope condition is that there exists some F such that

$$|f(\cdot)| \leq F(\cdot), \quad \forall f \in \mathcal{F}$$

and

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E} |F(w_i)|^{2+\kappa} < \infty \quad (8)$$

for some $\kappa > 0$.

Our primitive conditions are on the bracketing entropy of the function classes. For any $\varepsilon > 0$ and $p \in [2, \infty)$, the bracketing number $N_{[]}(\varepsilon, \mathcal{F}, L_p(P))$ is the smallest number N such that there exist functions $\{a_j, b_j\}_{j=1}^N$ such that for every d , some j satisfies

$$|f(\cdot, d) - a_j(\cdot)| \leq b_j(\cdot)$$

where

$$\max_j \sup_n \sup_{i \in \mathcal{I}_{2,n}} (\mathbb{E}[b_j(w_i)^p])^{1/p} \leq \varepsilon.$$

The bracketing entropy is

$$\mathcal{J}_{[]}(\mathcal{F}, L_p(P)) = \int_0^1 \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_p(P))} d\varepsilon.$$

D.1 Primitive conditions for stochastic equicontinuity

Let $w_i = (\epsilon_i, \nu_i, x_i, n)$. Note that x_i is fixed in our setup, and n is included to allow the log-likelihood to depend on the local parameter $\theta_n(h) = \theta_0 + h/\sqrt{n}$. Define the following log-likelihood quantities:

$$\begin{aligned} \ell_h(w_i, d) &= \sqrt{n} \log \frac{p_{y|z,x}(y_i(d) \mid z_i(d), x_i; \theta_n(h))}{p_{y|z,x}(y_i(d) \mid z_i(d), x_i; \theta_0)} \\ b_{n,h}(d) &= \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}_0[\ell_h(w_i, d)] \\ \Lambda_{2,n}^h(d) &= \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_{2,n}} \ell_h(w_i, d) = \mathbb{G}_n \ell_h(d) + b_{n,h}(d). \end{aligned}$$

In what follows, we refer to the following function classes:

$$\mathcal{F}_\psi^k = \{\psi_k(\cdot, d) : d \in \Delta\}, \quad k = 1, \dots, \ell, \quad \mathcal{F}_\ell^h = \{\ell_h(\cdot, d) : d \in \Delta\}.$$

Assumption 9 (Primitive conditions for equicontinuity): *1. For each k , the class*

$\mathcal{F}_\psi^k$ satisfies the Type IV condition of Andrews (1994) with $p = 2$. It has an envelope satisfying (8) for some $\kappa > 0$.

2. For every fixed h , the class $\mathcal{F}_\ell^h$ satisfies the Type IV condition of Andrews (1994) with $p = 2$. It has an envelope satisfying (8) for some $\kappa > 0$.

3. For every fixed h ,

$$\sup_{d \in \Delta} \left| b_{n,h}(d) + \frac{\rho}{2} h' J_2(d) h \right| \rightarrow 0$$

where J_2 is continuous.

Proposition 3: Suppose Assumptions 1, 2, 3, 5, and 9 hold. Then $\mathcal{J}_{\square}(\mathcal{F}_{\psi}^k, L_2(P_0)) < \infty$ for every k , and $\mathcal{J}_{\square}(\mathcal{F}_{\ell}^h, L_2(P_0)) < \infty$ for every h . Consequently, $A_{2,n}(\cdot)$ and $\Lambda_{2,n}^h(\cdot)$ are stochastically equicontinuous.

Proof. Theorem 5 of Andrews (1994) implies that $\mathcal{J}_{\square}(\mathcal{F}_{\psi}^k, L_2(P_0)) < \infty$ for every k . Combined with the envelope conditions, Theorem 4 of Andrews (1994) implies that $\mathbb{G}_n\psi_k(\cdot)$ is stochastically equicontinuous for each k with respect to the metric

$$\varrho(d, \tilde{d}) = \sup_n \sup_{i \in \mathcal{I}_{2,n}} (\mathbb{E}_0[|\psi_k(w_i, d) - \psi_k(w_i, \tilde{d})|^2])^{1/2}.$$

Since this is dominated by the Euclidean metric in the sense that $\varrho(d, \tilde{d}) \leq C_1 \|d - \tilde{d}\|^{C_2}$ for some constants C_1 and C_2 , $\mathbb{G}_n\psi_k(\cdot)$ is stochastically equicontinuous with respect to the Euclidean metric as well. Since the score is mean zero by DQM, $A_{2,n,k}(\cdot) = \mathbb{G}_n\psi_k(\cdot)$ and we conclude that the vector $A_{2,n}(\cdot)$ is stochastically equicontinuous.

The same argument applies to the log-likelihood to yield stochastic equicontinuity of $\mathbb{G}_n\ell_h(\cdot)$. By assumption, the centered drift term $b_{n,h}(\cdot)$ converges uniformly to a continuous function of d , so $\Lambda_{2,n}^h(\cdot) = \mathbb{G}_n\ell_h(\cdot) + b_{n,h}(\cdot)$ is stochastically equicontinuous as well. $\square$

D.2 Verifying primitive conditions for the Progres application

We first establish some regularity of the model described in Section 5. Let z_s denote the subsidy in grade s . Let $V_{\theta,\tau}(s, f; z)$ denote the ex ante value at age τ . In the terminal period,

$$V_{\theta,18}(s, f; z) = \gamma_9 \mathbb{1}[s \geq 10].$$

Choice-specific values are defined recursively by

$$V_{\theta,\tau}(s, f; z) = v_{\theta,\tau}^0(s, f; z) + \log\{1 + \exp(v_{\theta,\tau}(s, f; z))\}$$

where

$$\begin{aligned} v_{\theta,\tau}^0(s, f; z) &= \eta V_{\theta,\tau+1}(s, f; z), \\ v_{\theta,\tau}^1(s, f; z) &= \gamma_0 + \gamma_1\tau + \gamma_2s + \gamma_3f + \gamma_4\mathbb{1}[s \geq 7] \\ &\quad + \gamma_5(1-f)z_s + \gamma_6(1-f)\tau z_s + \gamma_7fz_s + \gamma_8f\tau z_s \end{aligned}$$

$$+ \eta [p_{\text{pass}}(\tau, s) V_{\theta, \tau+1}(s+1, f; z) + \{1 - p_{\text{pass}}(\tau, s)\} V_{\theta, \tau+1}(s, f; z)],$$

$$v_{\theta, \tau}(s, f; z) = v_{\theta, \tau}^1(s, f; z) - v_{\theta, \tau}^0(s, f; z).$$

In what follows, let x contain (τ, s, f) and define

$$v_{\theta}^y(z, x) = v_{\theta, \tau}^y(s, f; z),$$

$$v_{\theta}(z, x) = v_{\theta}^1(z, x) - v_{\theta}^0(z, x),$$

$$\varsigma_{\theta}(z, x) = p_{y|z, x}(1 | z, x; \theta) = \frac{\exp\{v_{\theta}(z, x)\}}{1 + \exp\{v_{\theta}(z, x)\}},$$

Proposition 4: *Suppose θ_0 is interior to a compact, convex $\bar{\Theta} \subseteq \Theta$. Suppose the covariate space, treatment space, and household state space are contained in compact and convex sets. Define*

$$M_{\theta} = \sup_{\theta, z, x} \|\nabla_{\theta} v_{\theta}(z, x)\|, \quad M_z = \sup_{\theta, z, x} \|\nabla_z v_{\theta}(z, x)\|,$$

$$M_{\theta z} = \sup_{\theta, z, x} \|\nabla_{z, \theta}^2 v_{\theta}(z, x)\|, \quad M_{\theta\theta} = \sup_{\theta, z, x} \|\nabla_{\theta, \theta}^2 v_{\theta}(z, x)\|,$$

$$L_{\psi} = M_{\theta z} + \frac{1}{4} M_{\theta} M_z$$

where θ ranges over $\bar{\Theta}$. Then these constants are finite and the following hold uniformly over $\theta \in \bar{\Theta}$:

1. Uniformly over y, z , and x , we have $\|\psi_{\theta}(y, z, x)\| \leq M_{\theta}$ and $\psi_{\theta}(y, z, x)$ is Lipschitz in y and z with Lipschitz constants M_{θ} and L_{ψ} .
2. ς_{θ_0} is Lipschitz in treatment with constant $\frac{1}{4} M_z$
3. For every $\ell_h \in \mathcal{F}_{\ell}^h$, $|\ell_h(\cdot, d)| \leq \|h\| M_{\theta}$ and ℓ_h is Lipschitz in y and z with Lipschitz constants $\|h\| M_{\theta}$ and $\|h\| L_{\psi}$.
4. For every fixed h , the log-likelihood satisfies the expansion

$$\sup_{z, x} \left| \mathbb{E}_0 \left[\log \frac{p_{y|z, x}(y | z, x; \theta_0 + h/\sqrt{n})}{p_{y|z, x}(y | z, x; \theta_0)} \mid z, x \right] \right. \\ \left. + \frac{1}{2n} h' \mathbb{E}_0 [\psi_{\theta_0}(y, z, x) \psi_{\theta_0}(y, z, x)' \mid z, x] h \right| = o(n^{-1}).$$

Proof. First, the terminal value and flow payoffs described in Section 5 are twice continuously differentiable in θ and z . Finite-horizon backward induction preserves this property, so $v_\theta(z, x)$ is twice continuously differentiable in θ and z . Compactness of the parameter, treatment, and covariate space implies that the derivatives of $v_\theta(z, x)$ are uniformly bounded, so that M_θ , M_z , $M_{\theta z}$, and $M_{\theta, \theta}$ are finite.

By direct calculation,

$$\psi_\theta(y, z, x) = \nabla_\theta \log p_{y|z,x}(y | z, x; \theta) = (y - \varsigma_\theta(z, x)) \nabla_\theta v_\theta(z, x)$$

and so the score is bounded uniformly by M_θ . Its gradient with respect to z satisfies

$$\begin{aligned} \|\nabla_z \psi_\theta(y, z, x)\| &\leq \|\nabla_z \varsigma_\theta(z, x)\| \|\nabla_\theta v_\theta(z, x)\| + |y - \varsigma_\theta(z, x)| \|\nabla_{z, \theta}^2 v_\theta(z, x)\| \\ &\leq \frac{1}{4} M_\theta M_z + M_{\theta z} = L_\psi \end{aligned}$$

and the change resulting from a change in y is

$$\|\psi_\theta(\tilde{y}, z, x) - \psi_\theta(y, z, x)\| = \|(\tilde{y} - y) \nabla_\theta v_\theta(z, x)\| \leq M_\theta |\tilde{y} - y|.$$

The gradient of ς_θ with respect to z is

$$\nabla_z \varsigma_\theta(z, x) = \varsigma_\theta(z, x)(1 - \varsigma_\theta(z, x)) \nabla_z v_\theta(z, x)$$

which is bounded by $\frac{1}{4} M_z$. For any $\ell_h \in \mathcal{F}_\ell^h$, the fundamental theorem of calculus implies that

$$\ell_h(w_i, d) = \int_0^1 h' \psi_{\theta_0 + th/\sqrt{n}}(y_i(d), z_i(d), x_i) dt$$

and so the envelope and increment bounds for ψ imply the same bounds for ℓ_h with constants multiplied by $\|h\|$.

We now verify the log-likelihood expansion. First, we have

$$\nabla_\theta \psi_\theta(y, z, x) = (y - \varsigma_\theta(z, x)) \nabla_{\theta, \theta}^2 v_\theta(z, x) - \varsigma_\theta(z, x)(1 - \varsigma_\theta(z, x)) \nabla_\theta v_\theta(z, x) \nabla_\theta v_\theta(z, x)'$$

and therefore $\sup_{\theta, y, z, x} \|\nabla_\theta \psi_\theta(y, z, x)\| \leq M_{\theta\theta} + \frac{1}{4} M_\theta^2$ and $\nabla_\theta \psi_\theta$ is integrable. Also, continuity of v_θ and compactness ensure that $\varsigma_\theta(z, x)$ is bounded away from 0 and 1,

so the log-likelihood ratio is integrable as well. Direct calculation yields the mean-zero property of the score

$$\mathbb{E}_0[\psi_{\theta_0}(y, z, x) \mid z, x] = \mathbb{E}_0[y - \varsigma_{\theta_0}(z, x) \mid z, x] \nabla_{\theta} v_{\theta_0}(z, x) = 0$$

and the information identity

$$\begin{aligned} \mathbb{E}_0[\nabla_{\theta} \psi_{\theta_0}(y, z, x) \mid z, x] &= -\varsigma_{\theta_0}(z, x)(1 - \varsigma_{\theta_0}(z, x)) \nabla_{\theta} v_{\theta_0}(z, x) \nabla_{\theta} v_{\theta_0}(z, x)' \\ &= -\mathbb{E}_0[\psi_{\theta_0}(y, z, x) \psi_{\theta_0}(y, z, x)' \mid z, x]. \end{aligned}$$

A Taylor expansion of the log-likelihood ratio then gives

$$\begin{aligned} &\log p_{y|z,x}(y \mid z, x; \theta_0 + h/\sqrt{n}) - \log p_{y|z,x}(y \mid z, x; \theta_0) \\ &= \frac{1}{\sqrt{n}} h' \psi_{\theta_0}(y, z, x) + \frac{1}{2n} h' \nabla_{\theta} \psi_{\theta_0}(y, z, x) h + o(n^{-1}). \end{aligned}$$

Since θ, z, x are all in compact sets and y is binary, $\nabla_{\theta} \psi_{\theta}(y, z, x)$ is uniformly continuous in θ , so the Taylor remainder above is uniformly $o(n^{-1})$ in z and x . Taking expectations conditional on z and x and using the information identity and the zero mean property of the score gives

$$\sup_{z,x} \left| \mathbb{E}_0 \left[\log \frac{p_{y|z,x}(y \mid z, x; \theta_0 + h/\sqrt{n})}{p_{y|z,x}(y \mid z, x; \theta_0)} \mid z, x \right] + \frac{1}{2n} h' \mathbb{E}_0[\psi_{\theta_0}(y, z, x) \psi_{\theta_0}(y, z, x)' \mid z, x] h \right| = o(n^{-1})$$

as desired. $\square$

We now verify that our model satisfies Assumption 9.

Proposition 5: *Maintain the assumptions of Proposition 4. Suppose the J -dimensional treatment is assigned by*

$$z_i(d) = \Phi'_0 x_i + \Phi'_1 x_i \mathbb{1}[\nu_i \leq G(\phi'_2 x_i)], \quad d = (\text{vec}(\Phi_0), \text{vec}(\Phi_1), \phi_2) \in \Delta$$

where $G : \mathbb{R} \mapsto [0, 1]$ is Lipschitz with constant K and $\nu_i \sim U[0, 1]$ conditional on x_i . Suppose ϵ_i and ν_i are independent conditional on x_i and Δ is compact.

Then for every finite $p \geq 2$ and every fixed h , $\mathcal{F}_{\psi}^k$ and $\mathcal{F}_{\ell}^h$ satisfy the Type IV condition (7) of Andrews (1994) for every $k = 1, \dots, \ell$. Both classes also have uniformly bounded envelopes satisfying (8) for some $\kappa > 0$.

If, in addition, Assumptions [1](#), [2](#), [3](#), and [5](#) hold, then for every fixed h , Assumption [9\(3\)](#) holds and consequently $A_{2,n}(\cdot)$ and $\Lambda_{2,n}^h(\cdot)$ are stochastically equicontinuous.

Proof. Let $I_i(d) = \mathbb{1}[\nu_i \leq G(\phi'_2 x_i)]$. If $\|\tilde{d} - d\| \leq r$, then

$$\|z_i(\tilde{d}) - z_i(d)\| \leq \|\tilde{\Phi}'_0 x_i - \Phi'_0 x_i\| + \|(\tilde{\Phi}'_1 - \Phi'_1) x_i\| + \|\Phi'_1 x_i\| \|I_i(\tilde{d}) - I_i(d)\|.$$

The first two terms are each bounded by $M_x r$ where $M_x = \sup_i \|x_i\|$. $\|\Phi'_1 x_i\|$ is bounded by a constant multiple of M_x as well by compactness of Δ .

To bound the last term, note that Lipschitz continuity of G implies that $|G(\tilde{\phi}'_2 x_i) - G(\phi'_2 x_i)| \leq K M_x r$ so that $I_i(\tilde{d}) \neq I_i(d)$ only if ν_i is within $K M_x r$ of $G(\phi'_2 x_i)$. Since ν_i is uniform, we have

$$P_0 \left(\sup_{\tilde{d}: \|\tilde{d} - d\| \leq r} |I_i(\tilde{d}) - I_i(d)| > 0 \mid x_i \right) \leq 2K M_x r.$$

Since the choice probability ς_{θ_0} is Lipschitz in z , on the event $S_i(d, r) = \{\sup_{\|\tilde{d} - d\| \leq r} |I_i(\tilde{d}) - I_i(d)| = 0\}$, we have $\sup_{\tilde{d}: \|\tilde{d} - d\| \leq r} \|z_i(\tilde{d}) - z_i(d)\| \leq C r$ for some constant C . Since the logistic distribution has a bounded density, this implies that there exists a C such that

$$P_0 \left(\sup_{\tilde{d}: \|\tilde{d} - d\| \leq r} |y_i(\tilde{d}) - y_i(d)| > 0 \cap S_i(d, r) \mid x_i \right) \leq C r.$$

Since $P_0(S_i(d, r)^c \mid x_i) \leq 2K M_x r$, we have

$$P_0 \left(\sup_{\tilde{d}: \|\tilde{d} - d\| \leq r} |I_i(\tilde{d}) - I_i(d)| + |y_i(\tilde{d}) - y_i(d)| > 0 \mid x_i \right) \leq C r$$

for some constant C .

We now use this fact to verify the Type IV condition for $\mathcal{F}_\psi^k$. We have

$$\begin{aligned} \psi(y_i(\tilde{d}), z_i(\tilde{d}), x_i) - \psi(y_i(d), z_i(d), x_i) &= [\psi(y_i(\tilde{d}), z_i(\tilde{d}), x_i) - \psi(y_i(d), z_i(\tilde{d}), x_i)] \\ &\quad + [\psi(y_i(d), z_i(\tilde{d}), x_i) - \psi(y_i(d), z_i(d), x_i)]. \end{aligned}$$

The first term is zero unless the outcome switches and is otherwise uniformly bounded

by Proposition 4. The second is Lipschitz in z_i by Proposition 4. Thus,

$$\mathbb{E}_0 \sup_{\tilde{d}: \|\tilde{d}-d\| \leq r} \|\psi(y_i(\tilde{d}), z_i(\tilde{d}), x_i) - \psi(y_i(d), z_i(d), x_i)\|^p \leq C_p r^p + C_p r \leq C_p r$$

and 7 holds for $\mathcal{F}_\psi^k$.

To verify the Type IV condition for $\mathcal{F}_\ell^h$, we apply the same decomposition to the log-likelihood. For any $\ell_h \in \mathcal{F}_\ell^h$, Proposition 4 implies that

$$|\ell_h(w_i, \tilde{d}) - \ell_h(w_i, d)| \leq \|h\| M_\theta |y_i(\tilde{d}) - y_i(d)| + \|h\| L_\psi \|z_i(\tilde{d}) - z_i(d)\|.$$

The first term is zero unless the outcome switches and is otherwise uniformly bounded since y_i is binary. On the event that the assignment indicator does not switch,

$$\sup_{\tilde{d}: \|\tilde{d}-d\| \leq r} \|z_i(\tilde{d}) - z_i(d)\| \leq Cr$$

and is otherwise uniformly bounded since z_i is bounded. Thus,

$$\mathbb{E}_0 \sup_{\tilde{d}: \|\tilde{d}-d\| \leq r} |\ell_h(w_i, \tilde{d}) - \ell_h(w_i, d)|^p \leq C_p r^p + C_p r \leq C_p r$$

and 7 holds for $\mathcal{F}_\ell^h$.

We now verify convergence of the drift term. Proposition 4 implies that

$$\begin{aligned} \sup_{z,x} \left| \mathbb{E}_0 \left[\log \frac{p_{y|z,x}(y \mid z, x; \theta_0 + h/\sqrt{n})}{p_{y|z,x}(y \mid z, x; \theta_0)} \mid z, x \right] \right. \\ \left. + \frac{1}{2n} h' \mathbb{E}_0 [\psi_{\theta_0}(y, z, x) \psi_{\theta_0}(y, z, x)' \mid z, x] h \right| = o(n^{-1}). \end{aligned}$$

Evaluate at $(z, x) = (z_i(d), x_i)$ and sum this across $i \in \mathcal{I}_{2,n}$ to obtain

$$\sup_{d \in \Delta} \left| b_{n,h}(d) + \frac{1}{2} h' \frac{n_2}{n} J_{2,n}(d) h \right| = o(1)$$

where

$$J_{2,n}(d) = \frac{1}{n_2} \sum_{i \in \mathcal{I}_{2,n}} \mathbb{E}_0 [\psi_{\theta_0}(y, z_i(d), x_i) \psi_{\theta_0}(y, z_i(d), x_i)'].$$

Table 1: Descriptive statistics for first-wave Progresa sample

	All	Boys		Girls	
		Educ < 6	Educ ≥ 6	Educ < 6	Educ ≥ 6
Observations	1,000	280	235	276	209
Age	12.56 (1.89)	11.70 (1.73)	13.78 (1.33)	11.38 (1.50)	13.89 (1.39)
Educ	4.99 (1.64)	3.78 (1.13)	6.51 (0.75)	3.84 (1.11)	6.43 (0.72)
Enrolled	0.806	0.889	0.757	0.899	0.627
Treatment assignment	0.627	0.586	0.660	0.623	0.651
Beneficiary	0.616	0.579	0.647	0.612	0.636
Beneficiary assigned	0.982	0.988	0.981	0.983	0.978

Notes: the first wave sample consists of 1000 observations drawn uniformly from the 12,031 observations in the analysis sample. All variables report the sample mean; continuous variables report sample standard deviations in parentheses. Treatment indicates initial experimental assignment, and beneficiary indicates realized treatment status.

The Type-IV condition for $\mathcal{F}_\psi^k$ and the uniform bound on the score imply that $J_{2,n}(d)$ is uniformly equicontinuous. Assumptions 1 and 5 imply that the pointwise limit is $J_2(d)$. Since Δ is compact, we conclude that this convergence is uniform in d . Hence

$$\sup_{d \in \Delta} \left| b_{n,h}(d) + \frac{\rho}{2} h' J_2(d) h \right| = o(1)$$

and Assumption 9(3) holds.

The envelope conditions are established in Proposition 4, so the result follows from Proposition 3. $\square$

E DETAILS OF THE PROGRESA APPLICATION

We construct the analysis sample from the Progresa raw data files contained in the replication package for Adhvaryu et al. (2023)⁷. We combine the October 1998 and March 1999 surveys and define enrollment in the 1998–99 school year as reported attendance in either wave. We merge the 1997 survey data to recover sex and other household characteristics. We retain children ages 10–17 in households classified

⁷For many years, the original Progresa data was made publicly available by the Mexican government, but the hosting website appears to have been taken down.

Table 2: Estimates of θ at each vertex and unrestricted fit

Parameter	Unrestricted (SE)	Boys Pri.	Boys Sec.	Girls Pri.	Girls Sec.
Constant	3.134 (0.367)	3.157	3.008	3.187	3.028
Female	-0.384 (0.082)	-0.337	-0.315	-0.401	-0.407
Enrollment Grade	0.88 (0.051)	0.838	0.831	0.839	0.829
Age	-0.728 (0.029)	-0.704	-0.687	-0.704	-0.683
Secondary School	-3.536 (0.213)	-3.336	-3.343	-3.331	-3.338
Subsidy Boys	5.108 (2.59)	6.652	8.852	2.926	6.261
Subsidy Girls	2.666 (2.245)	2.245	4.607	5.314	7.648
Subsidy Age Boys	-0.359 (0.264)	-0.649	-0.833	-0.281	-0.6
Subsidy Age Girls	-0.096 (0.238)	-0.219	-0.45	-0.529	-0.735
Terminal Completion	2.358 (0.417)	2.476	2.48	2.451	2.496
$\lambda_{\min}(H)$	–	0.0025	0.0105	0.0051	0.0245
LR Statistic	–	22.22	17.13	22.85	16.68
LR P-Value	–	< 0.0001	0.0007	< 0.0001	0.0008

Notes: All estimates are obtained from the holdout sample of 8,031 observations. Standard errors for the unrestricted model are reported in parentheses. $\lambda_{\min}(H)$ is the minimum eigenvalue of the negative Hessian of welfare at each vertex; positive values indicate strong concavity of welfare at the vertex. The likelihood ratio tests the weak complementarity constraints used to define each vertex. Subsidy amounts are in thousands of pesos, and age enters the model centered at 5.

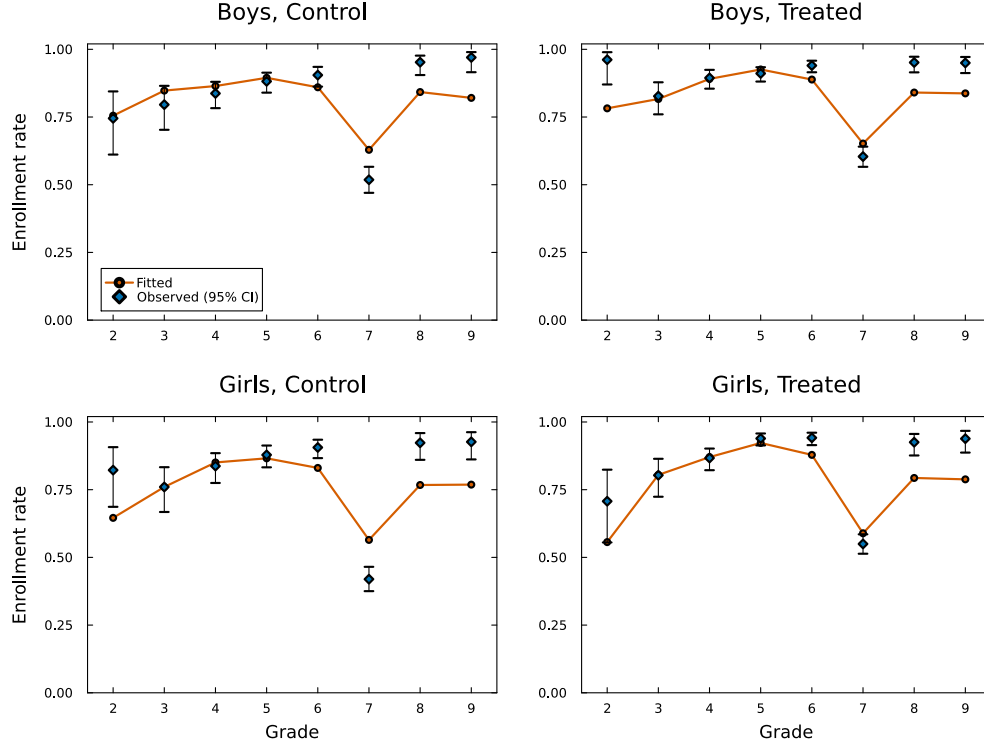
as poor and with fewer than nine years of completed schooling. These restrictions yield 12,031 children, from which we draw 1,000 children uniformly as the first-wave estimation sample. Table 1 reports descriptive statistics for this sample.

To calibrate the values of θ_0 used in our regret calculations, we estimate the model on a holdout set from the Progreso experiment. To capture uncertainty in the active set of constraints, we choose four values of θ_0 corresponding to the vertices of the feasible set which lie on the budget constraint. These vertices in policy coordinates are

$$\begin{array}{ll}
 (0.52875, 0, 0, 0) & (0, 0.705, 0, 0) \\
 (0, 0, 0.52875, 0) & (0, 0, 0, 0.705)
 \end{array}$$

where the entries correspond to the subsidy for boys in primary school, boys in secondary school, girls in primary school, and girls in secondary school, respectively. For each vertex, we choose θ_0 by estimating the model on the holdout set subject to the constraint that the optimal policy implied by θ is given by the vertex, and that the marginal effect of changing π at this vertex is equalized across the four groups. This

Figure 2: Enrollment-rate model fit



Note: Each panel shows predicted and observed enrollment rates for one of the four groups. The fitted enrollment rates use the estimates at the “Girls, Secondary” vertex of Table 2. Observed enrollment rates are plotted along with 95% Wilson confidence intervals. Both series are computed on the holdout sample of 8,031 observations.

means that at each vertex, the nonnegativity constraints bind weakly, so changing the subsidy in budget-neutral directions has no first-order effect on welfare.

Table 2 shows the unrestricted maximum-likelihood estimates of θ on the holdout set, as well as the estimates of θ at each vertex. We also report the minimum eigenvalue of H at each vertex to verify strong concavity of welfare at these points. We also conduct a likelihood ratio test of the weak complementarity constraints used to define each vertex. At the holdout sample size, all vertices can be rejected, but likelihood ratio tests fail to reject these points at the sample sizes considered in the Monte Carlo exercise. Figure 2 shows the predicted enrollment rates at the “Girls, Secondary” vertex compared to the observed enrollment rates in the holdout set.

For the Monte Carlo exercise, we sample from a Uniform ball of size K around each vertex θ_0 to generate draws of θ . K is chosen so that θ is difficult to distinguish

Table 3: Expected regret by experimental design

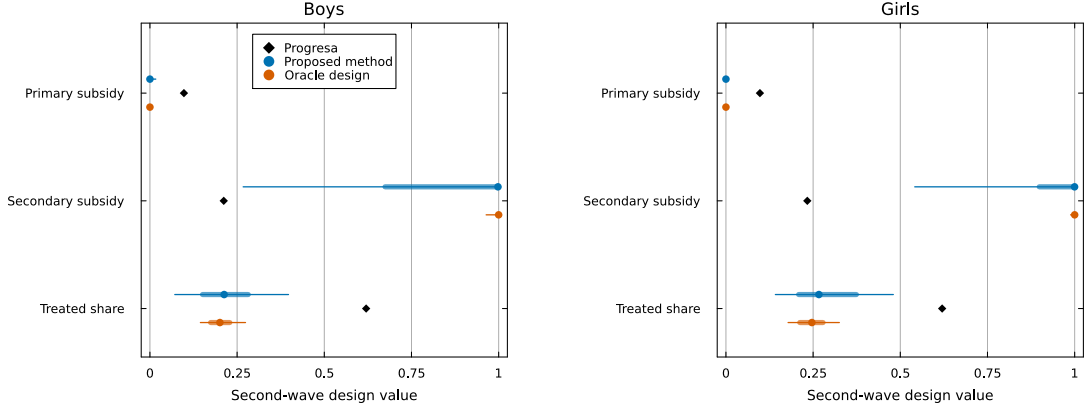
Sample Size	Pilot Only	Progresa	Proposed method	Oracle
<i>Panel A: Boys Primary</i>				
250	4.441 (0.242)	3.321 (0.204)	2.231 (0.158)	1.784 (0.139)
500	3.538 (0.211)	2.644 (0.187)	0.957 (0.093)	0.843 (0.085)
1000	2.327 (0.17)	1.305 (0.12)	0.403 (0.046)	0.402 (0.056)
<i>Panel B: Boys Secondary</i>				
250	4.366 (0.229)	3.292 (0.205)	2.329 (0.162)	1.877 (0.137)
500	3.342 (0.199)	2.544 (0.18)	1.066 (0.103)	1.039 (0.102)
1000	2.252 (0.166)	1.382 (0.117)	0.434 (0.046)	0.458 (0.054)
<i>Panel C: Girls Primary</i>				
250	4.528 (0.244)	3.396 (0.21)	2.474 (0.169)	1.972 (0.142)
500	3.481 (0.212)	2.637 (0.189)	1.036 (0.103)	0.9 (0.086)
1000	2.282 (0.166)	1.357 (0.12)	0.423 (0.047)	0.447 (0.06)
<i>Panel D: Girls Secondary</i>				
250	4.299 (0.233)	3.345 (0.204)	2.125 (0.148)	1.958 (0.138)
500	3.279 (0.203)	2.412 (0.175)	0.857 (0.079)	0.96 (0.09)
1000	2.038 (0.15)	1.199 (0.107)	0.473 (0.048)	0.462 (0.057)

Notes: Each panel reports expected regret in percentage points and Monte Carlo standard errors under a different reference point θ_0 , corresponding to a different vertex of the feasible set. Each vertex is characterized by a different group of children receiving a positive subsidy. Not all replications achieved convergence; expected regret is computed based on the $> 99\%$ of replications that converged for each vertex and sample size.

from θ_0 given the experimental data; it is 3.396 in Mahalanobis units. This has the interpretation that any θ in the ball lies in a 68% joint Wald confidence region around θ_0 with a sample size of 1000.

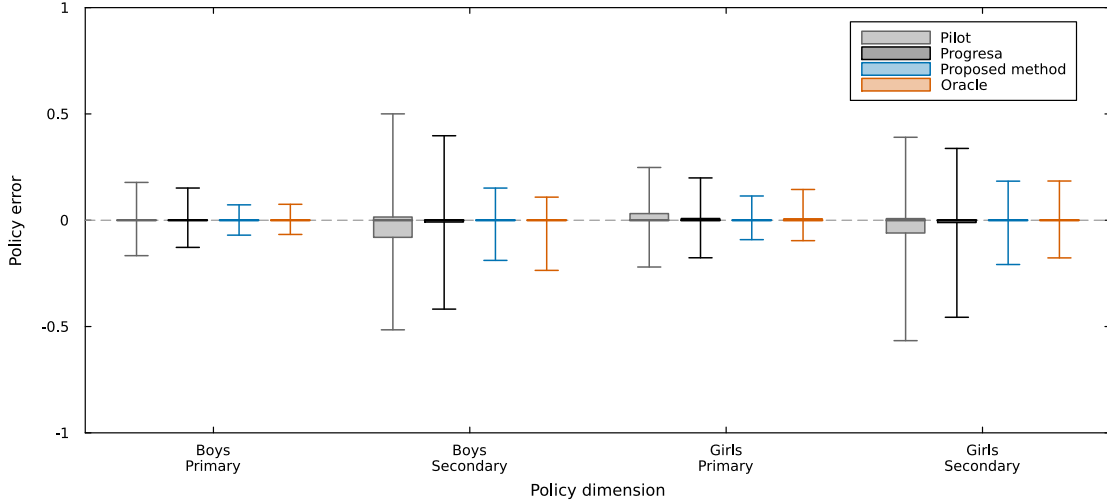
Table 3 reports the expected regret of the policy choice under different experimental designs for every vertex θ_0 and sample size. Figure 3 illustrates how the optimal experiment differs from the Progresa design, focusing on the case where θ_0 is at the “Girls Secondary” vertex and $n_1 = n_2 = 500$. Figure 4 likewise illustrates how the policy induced by the optimal design differs from the policy induced by the Progresa design by reducing the policy error $\hat{\pi}_{2,n} - \pi^*(\theta)$.

Figure 3: Variation in experimental designs



Note: The figure describes the second wave designs when θ_0 is at the “Girls Secondary” vertex and $n_1 = n_2 = 500$. The figure plots averages of the Progresa design over grades in primary/secondary school, but the simulations use the original Progresa design which varies by grade. The dot is the median, the thick line is the interquartile range, and the thin line is the 10th-90th percentile range. Subsidy amounts are in thousands of pesos. This figure is based on 498/500 replications for which the design optimization converged.

Figure 4: Policy errors by experimental design



Note: The figure reports the distribution of policy errors $\hat{\pi}_{2,n} - \pi^*(\theta)$ when $\hat{\pi}_{2,n}$ is estimated using different experimental designs. Results are shown for the case where θ_0 is at the “Girls Secondary” vertex and $n_1 = n_2 = 500$. The center line is the median, the box is the interquartile range, and the whiskers are the 10th-90th percentile range. Subsidy amounts are in thousands of pesos. This figure is based on 498/500 replications for which the design optimization converged.